Simulating non-unitary spin-triplet superconductivity and magnetism with impurities

by

Henry Sheehy

A thesis submitted in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

School of Physical Science
University of Kent

©Henry Sheehy, January 2nd, 2022

Abstract

Scanning tunnelling microscopy (STM), quasiparticle interference (QPI) and angle-resolved photoemission spectrocopy (ARPES) simulation of (topological) non-unitary spin-triplet unconventional superconductivity and magnetism, in quantum dots, thin films and 3D, with impurities.

Contents

1 Introduction **1**

2 Most general model of multi-orbital superconductivity with impurities **2**

2.1 Normal state: free electron gas 2

2.2 Superconducting state: electron-electron interaction 5

2.3 Self-consistent mean-field solution 6

2.4 Density functional theory via Green's functions 7

2.5 Experimental observables 11

2.5.1 Density of states 11

2.5.2 Linecut 12

2.5.3 Topography 12

2.5.4 Density of states map 12

2.5.5 Bogoliubov quasiparticle interference 12

2.5.6 Scanned Josephson Tunnelling Microscopy 13

2.5.7 Gap map 13

2.5.8 Cooper pair interference 13

3 Model of normal metal with impurities **14**

3.1 Analytical solution to an infinite, continuum model 14

3.1.1 Local density of states 14

3.1.2 Integrated local density of states 17

3.2 Applying the general model to the particular: benchmarking the normal state 18

3.2.1 Impurities modulate the local density of states (Friedel oscillations) 19

3.2.2 Density functional theory of impurities 19

3.2.3 Fourier transformation of the mean-field Hamiltonian 19

3.2.4 Quasiparticle interference 19

4 Models of multi-orbital superconductivity 43

4.1 Model of multi-orbital spin-singlet superconductivity 43

4.1.1 Model and reproducing known results 43

4.2 Model of multi-orbital (non-)unitary spin-triplet superconductor with impurities 43

5 Su-Schrieffer-Heeger model with multiorbital superconductivity 44

5.1 Su-Schrieffer-Heeger model with spin 44

5.2 Non-unitary spin-triplet Su-Schrieffer-Heeger model 45

6 Model of $LaNiGa_2$ 58

7 Two-dimensional Su-Schrieffer-Heeger model with multiorbital superconductivity 60

7.1 One-dimensional SSH with nearest-neighbour intraorbital hopping 61

7.2 Multiorbital tightbinding model, breaking chiral symmetry 61

7.3 Spin-chain in an external magnetic field 64

7.4 Non-unitary spin-triplet Su-Schrieffer-Heeger model 64

7.5 Type I extension 66

7.5.1 Single interchain hopping 66

8 Loop Josphson currents 67

9 Hubbard U simulator: manual 68

List of Tables

List of Figures

2.1 Simple multiorbital lattice 3

3.1 Energy dispersion of a two-dimensional metal described by a simple tight-binding model, such as Equation 2.1 with $s = \delta = 0$. The Fermi surface is projected onto the plane. Near the bottom of the band the Fermi surface is approximately quadratic. The Fermi surfaces corresponding to $\mu/t = -3.57$ and $\mu/t = -2.67$ are highlighted in white and pink, respectively. It is important to note that only the inner area, $k_x, k_y \in \{-\pi/2, \pi/2\}$ (thick dashed line) is observable via quasiparticle interference (QPI). 15

3.2 Top: The local density of states LDOS$(\mathbf{r}, \omega + i\epsilon)$ at r_n at zero bias ($\omega = 0$) with convergence factor $\epsilon = 0.1$ as a function of square systems of various sizes n with periodic boundary conditions. The system is the normal state ($\Delta = 0$), single band ($s = \delta = 0$) with chemical potential $\mu = -3.57$ and normalised by the hopping parameter ($t = 1.0$). The local density of states are homogeneous as there are no impurities. We observe the finite-size effects attenuate with increasing system size n. // Bottom: Memory stored in the matrix representation of the Hamiltonian and execution time of the simulation. 20

3.3 The local density of states LDOS($\mathbf{r}, \omega + i\epsilon$) along a line from $r_0 = [0, 0]$ to $r_{\text{diag}} = [(n-1)/2, 0]$ and $r_{\text{horiz}} = [(n-1)/2, (n-1)/2]$ at zero bias ($\omega = 0$) with convergence factor $\epsilon = 0.1$ for square systems of various sizes, as labelled in the legend, with periodic boundary conditions. The system is the normal state ($\Delta = 0$), single band ($s = \delta = 0$) with chemical potential $\mu = -3.57$, normalised by the hopping parameter ($t = 1.0$) and with an impurity at the centre of the lattice r_0 with coupling strength $V = 1.21$. The local density of states is plotted along a straight path from the centre of the lattice to half-way to the nearest neighbour, as a function of the distance in sites (top left) and as a function of the distance in sites normalised by the system size (top right); and similarly to half-way to the next-nearest neighbour (bottom left and right). We observe Friedel oscillations, that is to say, a modulation in the local density of states due to the translational symmetry breaking. 21

3.4 The local density of states at bias $\omega = 0$ and resolution $\epsilon = 0.1$ along a line from $[-44, 0]$ to $[44, 0]$ with an impurity located at $r_0 = [0, 0]$ with coupling strength $V/t = 1.21$ at various chemical potentials μ. The model is spinless and single band, hence superconductivity does not form. The numerical model consists of 79 sites along each dimension with periodic boundary conditions. The analytical result is the first-order expansion of the continuum Dyson equation with infinite boundary conditions (Equation 3.10) of a free-electron with an effective mass, an approximation which holds near the bottom of the band $\mu/t \gtrapprox -4$. The constant zeroth order term deviates from the exact numerical result as we move higher up the band, whilst the first order term reproduces the Friedel oscillations quantitatively. The non-self-consistent numerical result of the density is a stationary solution, as its degeneracy with the self-consistent solutions shows. The continuum result depends on a renormalisation of the potential which has not be taken into account and the free electron approximation. 22

3.5 The local density of states far away from the impurity does not depend strongly on the impurity coupling strength V/t. The model consists of 43×43 sites and chemical potential $\mu/t = -3.57$. 23

3.6 Quasiparticle interference (QPI) of the spinless, bandless normal state with $n = 79$ sites at zero applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.1$ The impurity, necessary for quasiparticle interference, is placed at $r = (0, 0)$ with coupling strength $V/t = 1.21$. The QPI interference pattern at zero bias probes the inner area $\{-\pi/2, \pi/2\}$ of the Fermi surface. 24

3.7 Quasiparticle interference (QPI) of a spinless, two-band normal state with $n = 57$ sites (maximum size on 64Gb RAM) at zero applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.1$. The bands have a rigid shift $s/t = 0.05$ and hybridisation $\delta/t = 0.075$. The impurity, necessary for quasiparticle interference, is placed at $r = (0, 0)$ with coupling strength $V/t = 1.21$. The QPI interference pattern at zero bias probes the inner area $\{-\pi/2, \pi/2\}$ of the Fermi surface. 25

3.8 The frequency of the Friedel oscillations in the local density of states increase as we move higher up the band with increasing μ/t. The number of sites along each dimension and the impurity coupling are fixed at 43 and $V/t = 1.21$, where we have normalised by the hopping. The bias and resolution are $\omega = 0$ and $\epsilon = 0.1$, respectively. 26

3.9 Spectrum with resolution $\epsilon = 0.1$ of the normal state to study the effect of an impurity located at the centre as a function of its coupling strength V/t. The model consists of 43×43 sites and chemical potential $\mu/t = -2.67$. . . 27

3.10 The local density of states in the normal state with chemical potential and coupling strengths $\mu/t = -3.57$ and $V/t = 1.21$, with 43 sites along each axis. Clockwise from top left: single central impurity; two impurities $(-3, 3)$ and $(3, -3)$; random arrangement of 20 impurities; and a *corral* of 15 impurities arranged in a circle with radius 14.5 sites. Top left figure, the arrow denote the two points probed in Figure 3.11. 28

3.11 The spectrum in the normal state, taken at the two points given by the arrows in Figure 3.10, top left frame: the impurity (navy) and five atoms away (green). The impurity potential and chemical potential are set to $V/t = 1.21$ and $\mu/t = -3.57$. In the next Figure 3.12 we study a linecut between these two points. 29

3.12 *The temples of Tamil Nadu.* A *linecut* is the local density of states at several equally spaced intervals along a straight line, moving away from the impurity. We see clearly the van Hove singularity as well as a peak due to the impurity. The impurity potential and chemical potential are set to $V/t = 1.21$ and $\mu/t = -3.57$. The line is shown in the top left frame of Figure 3.10, where the impurity (navy) and five atoms away (green) are given by the arrows. 30

3.13 The local density of states of the corral depicted three-dimensionally in order to clearly expose the standing waves in the electron density due to the ring of 15 impurities. The chemical potential is $\mu/t = -3.57$ and the coupling strength is $V/t = 1.21$, with 43 sites along each axis. The bias energy is $\omega/t = 0$ with resolution $\epsilon/t = 0.1$ 31

3.14 Self-consistent results for one band without impurities, chemical potential $\mu/t = -2.67$ and 5 sites along each axis. The critical temperature is in good agreement with the BCS conventional theory $T_c^{\mathrm{BCS}} = \Delta_0/1.76$. The ϕ-fields are degenerate and comparison less temperature dependant than the pairing potential Δ, as expected in conventional BCS theory. The absolute convergence factor is $\varepsilon = 10^{-3}$. 32

3.15 Results for $n = 25$. 33

3.16 Results for $n = 51$ with $\varepsilon = 10^{-2}$ order of magnitude less than in the $n = 5, 25$ simulations. 34

3.17 Self-consistent results for one band without impurities at zero temperature $T = 0$, with 5 sites along each axis. The ϕ-fields are degenerate. The absolute convergence factor is $\varepsilon = 10^{-3}$. 35

3.18 Results for $n = 25$ 36

3.19 Results for $n = 51$ with $\varepsilon = 10^{-2}$ order of magnitude less than in the $n = 5, 25$ simulations. 37

3.20 Self-consistent results for one band without impurities at zero temperature $T = 0$. The chemical potential is $\mu/t = -2.67$. The ϕ-fields are degenerate. The absolute convergence factor is $\varepsilon = 10^{-3}$. 38

3.21 Self-consistent results for one band with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$. 39

3.22 Self-consistent results for a multiorbital model with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$. . . . 40

3.23 Self-consistent results for a non-unitary triplet model with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$. 41

3.24 Quasiparticle interference (magnitude of the Fourier transform) of the normal state local density of states. Top left: Fourier transform of Figure 3.10, a single central impurity. Top right: 20 random impurities. Bottom: averaging over 8 and 100 simulations of 20 different random impurities. The chemical potential is $\mu/t = -3.57$ and the impurity coupling strength is $V/t = 1.21$. This local electron density is a function of the momentum of the Friedel waves and is 'double' the Fermi surface in the following sense $\boldsymbol{k}_{Fermi} = \frac{1}{2}\boldsymbol{k}_{Friedel}$. In the experimental literature, quasiparticle interference is performed by averaging over many trial experiments to remove noise from the image, as we simulate here. In our theoretical simulations however, we can use a single, central impurity to produce noiseless quasiparticle interference patterns. Such results differ only in magnitude, as we have shown here by using two different colourbars, one for the single impurity and another for the three examples of many-impurity averaging. 42

5.1 SSH unit cell 46

5.2 Normal SSH phase diagram with magnetism, staggered density and IPR. . . 47

5.3 Normal SSH LDOS 48

5.4 Normal SSH LDOS 49
5.5 Normal SSH LDOS two impurities 50
5.6 Building the spin-tripelt SSH phase diagram 52
5.7 Spin-triplet SSH phase diagram without impurity 53
5.8 Spin-triplet SSH LDOS 54
5.9 Spin-triplet SSH magnetism 55
5.10 Spin-triplet SSH staggered density 56
5.11 Spin-triplet SSH IPR 57

7.1 fig:ssh-one$_d im$ 62
7.2 fig:multiorbital$_c hain$ 63

Notation

a	edge length of a cell
d	cell-cell distance
θ	inter-cell angle
q	compensation charge
μ	chemical potential
T	temperature
t	hopping parameter
U	on-site Coulomb repulsion
V_1	nearest-neighbour Colomb repulsion ($V_1 = 1/a$)
V_0	next-nearest-neighbour Colomb repulsion ($V_0 = 1/\sqrt{2a^2}$)
$V_{1\mathrm{crit}}$	mean field critical V_1
ΔE_{S}	singlet-triplet splitting
P_k	cell polarization (of cell k)
P_{D}	polarization of the driver cell
ϵ_k^{I}	relative error of the polarization of cell k for the Ising model
ϵ_k^{B}	relative error of the polarization of cell k for the bond model
χ_{kl}	linear polarization response of cell k with respect to cell l
χ	linear nearest-neighbour polarization response (the same for all cells)
χ_{e}	linear polarization response of the system to an external driver cell
H_k^{c}	single-cell Hamiltonian
H_{kl}^{cc}	Coulombic cell-cell interaction (between cells k and l)
S_k^z	pseudo spin (of cell k)
J_{kl}	Ising cell-cell interaction (between cells k and l)
J'_{kl}	modified Ising cell-cell interaction (between cells k and l)

γ	Ising transverse field / effective hopping parameter

Chapter 1

Introduction

Chapter 2

Most general model of multi-orbital superconductivity with impurities

We describe a general theory of multi-orbital superconductivity with unconventional pairing on a 2D lattice with periodic boundary conditions. The lattice is doped with impurities, which may, for example, be charge or magnetic. The non-interacting part of the model is a 2D electron gas with spin channels σ, orbital channels m, described by an on-site tensor $\boldsymbol{\mu}$, a nearest-neighbour hopping tensor $\boldsymbol{t}$ and a tensor $\boldsymbol{V}$ to dope the lattice with impurities. We describe electron-electron interactions via an unconventional BCS mean-field theory. The interacting model is solved via a generalised *Bogoliubov-de Gennes* self-consistent mean-field calculation.

Impurities in the quasipariticle sea give rise to density modulations in the local density of states (LDOS) known as Friedel oscillations. The Friedel oscillations -as observed via scanning tunnelling microscopy -characterise the dispersion relation, lifetime and pairing of quasiparticles, which give us a measure to test the theory.

2.1 Normal state: free electron gas

The non-interacting part of the model is a description of the normal state, that is, a generalised tight-binding model of a metal. The most general model of nearest-neighbour

../../05-fig/01-01-simple-lattice.pdf

Figure 2.1:

interaction is given by

$$\hat{\mathcal{H}}_0 = \sum_{\mathbf{i}\in\Omega} \sum_{\sigma\in\{\uparrow,\downarrow\}} \sum_{m\in\{+,-\}} \left[(\boldsymbol{\mu}_{\sigma,\tau,m,n} + \boldsymbol{V}_{\mathbf{i},\sigma,\tau,m,n}) \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \hat{c}_{\mathbf{i},\tau,n} \right] + \sum_{\mathbf{i}\in\Omega} \sum_{\mathbf{j}\in\{\hat{\mathbf{x}},\hat{\mathbf{y}}\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \sum_{m,n\in\{+,-\}} \left[\boldsymbol{t}_{\sigma,\tau,m,n} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \hat{c}_{\mathbf{i}+\mathbf{j},\tau,n} + \text{h.c.} \right] \quad (2.1)$$

where Ω is a set of N lattice points with periodic boundary conditions, labelled $\mathbf{i} = (i_x, i_y)$. Let us write the onsite tensor $\boldsymbol{\mu}$ in terms of its spin and orbital sectors

$$\boldsymbol{\mu} = \boldsymbol{\mu}_{\text{spin}} \otimes \boldsymbol{\mu}_{\text{orbital}}. \quad (2.2)$$

For a general solid with a chemical potential μ and two orbitals with a rigid energy shift s and hybridisation factor δ, the orbital sector of the onsite tensor would be

$$\boldsymbol{\mu}_{\text{spin}} = -\mu\mathbb{1} - s\sigma_z + \delta\sigma_x. \quad (2.3)$$

In the simplest model, the spin component would be trivial

$$\boldsymbol{\mu}_{\text{orbital}} = \mathbb{1}. \quad (2.4)$$

The nearest-neighbour hopping tensor can be resolved into spin and orbital sectors as

$$\boldsymbol{T} = \boldsymbol{t}_{\text{spin}} \otimes \boldsymbol{t}_{\text{orbital}}, \quad (2.5)$$

where the spin sector is trivial in the simplest theory

$$\boldsymbol{T}_{\text{spin}} = \mathbb{1}. \quad (2.6)$$

The orbital sector would also be trivial for the simplest theory

$$\boldsymbol{T}_{\text{orbital}} = -t\mathbb{1}. \quad (2.7)$$

To include inter-orbital hopping we would include an inter-orbital hopping term t_{orbital} into the hopping tensors as

$$\boldsymbol{T} = \mathbb{1} \otimes \left(-t\mathbb{1} - t_{\text{orbital}}\sigma_x\right). \quad (2.8)$$

Impurities enter the model via a subset $\Omega' \subset \Omega$ of $M < N$ lattice points randomly chosen from Ω

$$\boldsymbol{V}_{\mathbf{i}} = \begin{cases} \boldsymbol{V} & \text{if } \mathbf{i} \in \Omega', \\ 0 & \text{otherwise,} \end{cases} \tag{2.9}$$

where for nonmagnetic impurities

$$\boldsymbol{V} = -V\mathbb{1}. \tag{2.10}$$

Magnetic impurities can be modelled with a magnetic bias b which enters as

$$\boldsymbol{V} = (-V\mathbb{1} + b\sigma_z) \otimes \mathbb{1}, \tag{2.11}$$

where the trivial component refers to the orbital sector. A similar equation in the orbital sector models orbital-biased impurities.

Include cartoon of model

2.2 Superconducting state: electron-electron interaction

Superconductivity arises due to electron-electron attraction. We consider the most general unconventional theory of on-site interaction. The theory is a generalisation of the BCS theory. The Hamiltonian describing this model is

$$\hat{\mathcal{H}}_I = \sum_{\mathbf{i}\in\Omega} \sum_{\sigma\in\{\uparrow,\downarrow\}} \sum_{m\in\{+,-\}} \boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau} \hat{c}^\dagger_{\mathbf{i},\sigma,m} \hat{c}^\dagger_{\mathbf{i},\tau,n} \hat{c}_{\mathbf{i},\tau,n} \hat{c}_{\mathbf{i},\sigma,m}. \tag{2.12}$$

In the simplest theory of homogenous, interorbital, equal-spin pairing Ref. [**?**] the interaction tensor $\boldsymbol{U}$ is given by

$$\boldsymbol{U} = -U\mathbb{1} \otimes \sigma_x \tag{2.13}$$

with $U > 0$ being the effective attractive interaction necessary for superconductivity[1]. The first sector corresponds to spin and the second to orbital. This is mathematically similar to the original BCS theory of on-site, opposite spin pairing (with only one orbital), except with the orbital indices playing the role of the of the BCS spin indices.

$$\boldsymbol{U}_{\text{BCS}} = -U\sigma_x \otimes \mathbb{1} \tag{2.14}$$

[1] $U < 0$ corresponds to an effective Coulomb repulsion strength, which may be interesting to model other phenomena such as name phenomena.

Together, the normal state and interacting Hamiltonians form the unconventional *Bogoliubov-de Gennes* Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{BdG}} = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}_I. \tag{2.15}$$

In order to obtain observables, the Hamiltonian must be diagonalised. The usual way to do this is via a variational mean-field theory, which involves minimising the free energy, given by

$$F \equiv \langle H \rangle - TS = -k_B T \ln Z \tag{2.16}$$

where Z is the partition function for the system, T is the temperature, S is the entropy and k_B is Boltzmann's constant.

2.3 Self-consistent mean-field solution

The mean-field theory introduces a new Hamiltonian with variable parameters which can be solved exactly. The formal details are given in many places. Instead, we introduce mean-field theory as the *art* of introducing the minimum necessary mean-fields, either by necessity in order for a model to exhibit e.g. magnetism (spin-density fields) or in order to explore a wider phase space to find a free-energy minimum. In our model we include a Gor'kov-like pairing mean-field tensor $\boldsymbol{\Delta}^{mn}_{\mathbf{i}\sigma\tau}$, whose nonzero value is required for superconductivity. We also include a Hartree-like tensor $\boldsymbol{\eta}^{m}_{\mathbf{i}\sigma}$ which describes renormalisations of charge-spin-orbital density fields. Fock-like terms are not our main-interest, but for generality we include them as $\boldsymbol{\phi}^{mn}_{\mathbf{i}\sigma\tau}$. These are all the terms expected from Wick's theorem. The mean-field interaction for our model is

$$\begin{aligned}\hat{\mathcal{H}}_I^{\mathrm{MF}} = \sum_{\mathbf{i}\in\Omega} \sum_{\sigma\in\{\uparrow,\downarrow\}} \sum_{m\in\{+,-\}} &\left[\boldsymbol{\eta}^{m}_{\mathbf{i}\sigma} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \hat{c}_{\mathbf{i},\sigma,m} + \bar{\boldsymbol{\eta}}^{m}_{\mathbf{i}\sigma} \hat{c}_{\mathbf{i},\sigma,m} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m}\right] \\ + \sum_{\mathbf{i}\in\Omega} \sum_{\sigma,\tau\in\{\uparrow,\downarrow\}} \sum_{m,n\in\{+,-\}} &\Big[\boldsymbol{\phi}^{mn}_{\mathbf{i}\sigma\tau} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \hat{c}_{\mathbf{i},\tau,n} + \bar{\boldsymbol{\phi}}^{mn}_{\mathbf{i}\sigma\tau} \hat{c}_{\mathbf{i},\tau,n} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \\ &+ \boldsymbol{\Delta}^{mn}_{\mathbf{i}\sigma\tau} \hat{c}^{\dagger}_{\mathbf{i},\sigma,m} \hat{c}^{\dagger}_{\mathbf{i},\tau,n} + \bar{\boldsymbol{\Delta}}^{mn}_{\mathbf{i}\sigma\tau} \hat{c}_{\mathbf{i},\tau,n} \hat{c}_{\mathbf{i},\sigma,m}\Big]\end{aligned} \tag{2.17}$$

The most general self-consistency equations which minimise[2] the free energy are formally given by

$$\begin{aligned}
\boldsymbol{\eta}^{m}_{\mathbf{i}\sigma} &= -\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle \hat{c}^{\dagger}_{\mathbf{i},\tau,n}\hat{c}_{\mathbf{i},\tau,n}\rangle && (2.18)\\
\bar{\boldsymbol{\eta}}^{m}_{\mathbf{i}\sigma} &= -\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle c_{\mathbf{i},\tau,n}\hat{c}^{\dagger}_{\mathbf{i},\tau,n}\rangle && (2.19)\\
\boldsymbol{\phi}^{mn}_{\mathbf{i}\sigma\tau} &= +\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle \hat{c}^{\dagger}_{\mathbf{i},\tau,n}\hat{c}_{\mathbf{i},\sigma,m}\rangle && (2.20)\\
\bar{\boldsymbol{\phi}}^{mn}_{\mathbf{i}\sigma\tau} &= -\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle \hat{c}^{\dagger}_{\mathbf{i},\sigma,m}\hat{c}_{\mathbf{i},\tau,n}\rangle && (2.21)\\
\boldsymbol{\Delta}^{mn}_{\mathbf{i}\sigma\tau} &= -\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle \hat{c}_{\mathbf{i},\tau,n}\hat{c}_{\mathbf{i},\sigma,m}\rangle && (2.22)\\
\bar{\boldsymbol{\Delta}}^{mn}_{\mathbf{i}\sigma\tau} &= -\boldsymbol{U}^{m,n}_{\mathbf{i},\sigma,\tau}\langle \hat{c}^{\dagger}_{\mathbf{i},\sigma,m}c^{\dagger}_{\mathbf{i},\tau,n}\rangle && (2.23)
\end{aligned}$$

In general, $\bar{\boldsymbol{\eta}}$, $\bar{\boldsymbol{\phi}}$, $\bar{\boldsymbol{\Delta}}$ may be arbitrary. However, symmetry forces $\bar{\boldsymbol{\eta}} = -\boldsymbol{\eta}$, $\bar{\boldsymbol{\phi}} = -\boldsymbol{\phi}^*$ and $\bar{\boldsymbol{\Delta}} = -\boldsymbol{\Delta}^*$.

The mean-field theory of low energy excitations is then given by the following Bogoliubov-de Gennes (BdG) effective Hamiltonian

$$\hat{\mathcal{H}}^{\mathrm{MF}}_{\mathrm{BdG}} = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}^{\mathrm{MF}}_{I} + \mathcal{E}_{\mathrm{gs}}, \tag{2.24}$$

where $\mathcal{E}_{\mathrm{gs}}$ is calculate this. Solutions of the BdG mean-field Hamiltonian are given by

$$\hat{\mathcal{H}}^{\mathrm{MF}}_{\mathrm{BdG}}\Psi_n = \mathcal{E}_n\Psi_n. \tag{2.25}$$

Let us note that the particle-hole symmetry of the Bogoliubov-de Gennes Hamiltonian on its solutions. If we write a spinor Ψ_n with eigenvalue $\mathcal{E}_n > 0$ as $\Psi_n = ((u_{im\sigma})_n, (v_{im\sigma})_n)$, then by the particle-hole symmetry, the solution $\bar{\Psi}_n$ with $\mathcal{E}_n < 0$ is $\bar{\Psi}_n = ((-v^*_{\mathbf{i}m\sigma})_n, (u^*_{\mathbf{i}m\sigma})_n)$.

Write the diagonalised Hamiltonian as

$$\hat{\mathcal{H}}^{\mathrm{MF}}_{\mathrm{BdG}} = \mathcal{E}_{\mathrm{gs}} + \sum_n \mathcal{E}_n\hat{\gamma}^{\dagger}_n\hat{\gamma}_n - \mathcal{E}_n\hat{\gamma}_n\hat{\gamma}^{\dagger}_n, \tag{2.26}$$

then

2.4 Density functional theory via Green's functions

Read https://www.nobelprize.org/uploads/2018/06/kohn-lecture.pdf Let the solutions to

[2] To be precise, these are stationary values not necessarily the minimum.

the *Bogoliubov de-Gennes* Hamiltonian be written as $(\boldsymbol{v}_{\mathbf{i},\sigma,m})_n$ with energy E_n, where the indices label the sites $\mathbf{i} = (i_x, i_y)$, spin s and orbital m numbers. The solutions are related by particle-hole symmetry. Indeed, write the non-unitary time-reversal operator as

$$\hat{\mathcal{T}} \equiv \hat{\mathcal{C}} \left(-i\sigma_y\right), \tag{2.27}$$

where $\hat{\mathcal{C}}$ is the complex conjugation operator. Then for every solution $\boldsymbol{\Psi}_n$ with energy $\mathcal{E}_n > 0$, there exists a solution $\bar{\boldsymbol{\Psi}} = \hat{\mathcal{T}}\boldsymbol{\Psi}_n$ with energy $-\mathcal{E}_n$. The *Nambu-Gor'kov Green*'s function $\hat{\boldsymbol{\mathcal{G}}}$ is the kernel to the *Bogoliubov-de Gennes* Hamiltonian $\hat{\mathcal{H}}$, formally written

$$\hat{\boldsymbol{\mathcal{G}}}(z) \equiv \left(z - \hat{\mathcal{H}}\right)^{-1}, \tag{2.28}$$

where $z \in \mathbb{C}$. In the eigenbasis $\boldsymbol{\Psi}_n$, $\mathcal{E}_n$ of the Hamiltonian, the Green's function becomes

$$\boldsymbol{\mathcal{G}}(z) = \sum_n \frac{\boldsymbol{\Psi}_n \boldsymbol{\Psi}_n^\dagger}{z - \mathcal{E}_n} + \frac{\bar{\boldsymbol{\Psi}}_n \bar{\boldsymbol{\Psi}}_n^\dagger}{z + \mathcal{E}_n}. \tag{2.29}$$

We will now study some properties of this Green's function. Given a positive solution $\boldsymbol{\Psi}_n = (u_{\mathbf{i},\sigma,m})_n, (v_{\mathbf{i},\sigma,m})_n)$, its corresponding negative energy solution is $\bar{\boldsymbol{\Psi}}_n = \left(-v^*_{\mathbf{i},\sigma,m})_n, (u^*_{\mathbf{i},\sigma,m})_n\right)$. Then using this particle-hole symmetry, the Green's function can be written as

$$\begin{aligned}\boldsymbol{\mathcal{G}}^{m,m'}_{\sigma,\sigma'}\left(z;\mathbf{r},\mathbf{r}'\right) = \sum_n \begin{pmatrix} (u_{\mathbf{i}',\sigma',m'})_n \\ (v_{\mathbf{i}',\sigma','m'})_n \end{pmatrix} \frac{1}{z - \mathcal{E}_n} \begin{pmatrix} (u^*_{\mathbf{i},\sigma,m})_n & (v^*_{\mathbf{i},\sigma,m})_n \end{pmatrix} \\ + \sum_n \begin{pmatrix} (-v^*_{\mathbf{i}',\sigma',m'})_n \\ (u^*_{\mathbf{i}',\sigma',m'})_n \end{pmatrix} \frac{1}{z + \mathcal{E}_n} \begin{pmatrix} (-v_{\mathbf{i},\sigma,m})_n & (u_{\mathbf{i},\sigma,m})_n \end{pmatrix},\end{aligned} \tag{2.30}$$

where we have used $\mathbf{r}$ and $\mathbf{i}$ interchangeably since they are in one-to-one correspondence. This Green's function has can be split into normal and anomalous components

$$\boldsymbol{\mathcal{G}}\left(z\right) = \begin{pmatrix} \mathcal{G}\left(z\right) & \mathcal{F}\left(z\right) \\ \bar{\mathcal{F}}\left(z\right) & \bar{\mathcal{G}}\left(z\right) \end{pmatrix}. \tag{2.31}$$

Then the noninteracting electron-electron Green's function matrix $\mathcal{G}$ can be read from this

equation as

$$\mathcal{G}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)=\sum_{n}\left[\frac{(u_{\mathbf{i}',\sigma',m'})_n(u_{\mathbf{i},\sigma,m}^*)_n}{z-\mathcal{E}_n}+\frac{(v_{\mathbf{i}',\sigma',m'}^*)_n(v_{\mathbf{i},\sigma,m})_n}{z+\mathcal{E}_n}\right]. \tag{2.32}$$

The hole-hole component is

$$\bar{\mathcal{G}}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)=\sum_{n}\left[\frac{(v_{\mathbf{i}',\sigma',m'})_n(v_{\mathbf{i},\sigma,m}^*)_n}{z-\mathcal{E}_n}+\frac{(u_{\mathbf{i}',\sigma',m'}^*)_n(u_{\mathbf{i},\sigma,m})_n}{z+\mathcal{E}_n}\right]. \tag{2.33}$$

Similar expressions exist for the anomalous Green's functions

$$\mathcal{F}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)=\sum_{n}\left[\frac{(u_{\mathbf{i}',\sigma',m'})_n(v_{\mathbf{i},\sigma,m}^*)_n}{z-\mathcal{E}_n}+\frac{(-v_{\mathbf{i}',\sigma',m'}^*)_n(u_{\mathbf{i},\sigma,m})_n}{z+\mathcal{E}_n}\right] \tag{2.34}$$

and

$$\bar{\mathcal{F}}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)=\sum_{n}\left[\frac{(v_{\mathbf{i}',\sigma',m'})_n(u_{\mathbf{i},\sigma,m}^*)_n}{z-\mathcal{E}_n}+\frac{(-u_{\mathbf{i}',\sigma',m'}^*)_n(v_{\mathbf{i},\sigma,m})_n}{z+\mathcal{E}_n}\right] \tag{2.35}$$

The kernels are normalised so that they trace to unity $\mathrm{Tr}\equiv 1$. It is useful to note the algebraic relations

$$\left(\frac{1}{z}\right)^*=\frac{1}{z^*},\quad (wz)^*=z^*w^* \tag{2.36}$$

for any complex numbers w,z. Denote the transpose operation on the Green's function as

$$\left[\mathcal{G}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)\right]^{\mathrm{T}}\equiv\mathcal{G}_{\sigma',\sigma}^{m',m}\left(z;\mathbf{r}',\mathbf{r}\right). \tag{2.37}$$

Then we note some properties of the generalised Nambu-Gor'kov Green's function which follow. Let $\mathcal{G}\left(z\right)\equiv\mathcal{G}_{\sigma,\sigma'}^{m,m'}\left(z;\mathbf{r},\mathbf{r}'\right)$. Then

$$\mathcal{G}\left(z\right)=\left[\mathcal{G}\left(z^*\right)\right]^{\dagger}=-\left[\bar{\mathcal{G}}\left(-z\right)\right]^{\mathrm{T}}=-\left[\bar{\mathcal{G}}\left(-z^*\right)\right]^{*} \tag{2.38}$$

and

$$\mathcal{F}\left(z\right)=\left[\mathcal{F}\left(-z\right)\right]^{\mathrm{T}}=\left[\bar{\mathcal{F}}\left(z^*\right)\right]^{\dagger}. \tag{2.39}$$

The Green's function matrix then becomes

$$\boldsymbol{\mathcal{G}}\left(z\right)=\begin{pmatrix}\mathcal{G}\left(z\right) & \mathcal{F}\left(z\right)\\ \left[\mathcal{F}\left(z^{*}\right)\right]^{\dagger} & -\left[\mathcal{G}\left(-z\right)\right]^{\mathrm{T}}\end{pmatrix}. \tag{2.40}$$

A useful spectral property of Green's functions is that

$$\Im\boldsymbol{\mathcal{G}}(z^{*})=-\Im\boldsymbol{\mathcal{G}}(z),\quad \Im\boldsymbol{\mathcal{G}}^{R}(\omega)=-\Im\boldsymbol{\mathcal{G}}^{A}(\omega), \tag{2.41}$$

where $\boldsymbol{\mathcal{G}}^{R}(\omega)$ and $\boldsymbol{\mathcal{G}}^{A}(\omega)$ are the *retarded* and *advanced* Green's functions, defined by a convergence factor $\epsilon>0$ and $\epsilon<0$, respectively. We define the electron density of states as

$$\mathrm{DOS}_{\sigma,\tau}^{m,n}\left(\omega,\mathbf{r}\right)\equiv-\frac{1}{\pi}\Im\mathcal{G}_{\sigma,\tau}^{R\,m,n}\left(\omega;\mathbf{r},\mathbf{r}\right), \tag{2.42}$$

Similarly, the anomalous density of states is

$$\mathrm{ADOS}_{\sigma,\tau}^{m,n}\left(\omega,\mathbf{r}\right)\equiv-\frac{1}{\pi}\Im\mathcal{F}_{\sigma,\tau}^{R\,m,n}\left(\omega;\mathbf{r},\mathbf{r}\right), \tag{2.43}$$

In order to carry out the self-consistent calculation, we must obtain the normal and anomalous densities. These are

$$\begin{aligned}\rho_{\sigma,\sigma'}^{m,m'}\left(\mathbf{r}\right) &\equiv \int_{\omega_B}^{\omega_F}\mathrm{d}\omega\left[\Im\mathcal{G}_{\sigma,\sigma'}^{R\,m,m'}\left(\omega;\mathbf{r},\mathbf{r}\right)f(\omega)+\Im\bar{\mathcal{G}}_{\sigma,\sigma'}^{R\,m,m'}\left(\omega;\mathbf{r},\mathbf{r}\right)f(-\omega)\right] && (2.44)\\ &= \sum_{n}\left[(u_{\mathbf{i}',\sigma',m'})_n(u^{*}_{\mathbf{i},\sigma,m})_n f_n+(v^{*}_{\mathbf{i}',\sigma',m'})_n(v_{\mathbf{i},\sigma,m})_n(1-f_n)\right]. && (2.45)\end{aligned}$$

and

$$\begin{aligned}\chi_{\sigma,\sigma'}^{m,m'}\left(\mathbf{r}\right) &\equiv \frac{1}{2}\int_{\omega_B}^{\omega_F}\mathrm{d}\omega\left[\Im\mathcal{F}_{\sigma,\sigma'}^{R\,m,m'}\left(\omega;\mathbf{r},\mathbf{r}\right)(1-2f(\omega))+\Im\bar{\mathcal{F}}_{\sigma,\sigma'}^{R\,m,m'}\left(\omega;\mathbf{r},\mathbf{r}\right)(1-2f(-\omega))\right] && (2.46)\\ &= \frac{1}{2}\sum_{n}\left[(u_{\mathbf{i}',\sigma',m'})_n(v^{*}_{\mathbf{i},\sigma,m})_n(1-2f_n)+(-v^{*}_{\mathbf{i}',\sigma',m'})_n(u_{\mathbf{i},\sigma,m})_n(2f_n-1)\right], && (2.47)\end{aligned}$$

where ω_B is the bottom of the band and f is the Fermi function. Using the above properties

of the Green's functions, at zero temperature $T \to 0$ these become

$$\rho_{\sigma,\sigma'}^{m,m'}(\mathbf{r}) = -\frac{1}{\pi}\int_{\omega_B}^{\omega_F} \mathrm{d}\omega \Im \mathcal{G}_{\sigma,\sigma'}^{R\,m,m'}(\omega;\mathbf{r},\mathbf{r}) \tag{2.48}$$

$$= \sum_n \left[(u_{\mathbf{i}',\sigma',m'})_n (u^*_{\mathbf{i},\sigma,m})_n f_n\right] \tag{2.49}$$

and

$$\chi_{\sigma,\sigma'}^{m,m'}(\mathbf{r}) = -\frac{1}{2\pi}\int_{\omega_B}^{\omega_F} \mathrm{d}\omega \Im \left[\mathcal{F}_{\sigma,\sigma'}^{R\,m,m'}(\omega;\mathbf{r},\mathbf{r}) + \left[\mathcal{F}_{\sigma,\sigma'}^{A\,m,m'}(-\omega;\mathbf{r},\mathbf{r})\right]^{\mathrm{T}}\right] \tag{2.50}$$

$$= \frac{1}{2}\sum_n \left[(u_{\mathbf{i}',\sigma',m'})_n (v^*_{\mathbf{i},\sigma,m})_n - (-v^*_{\mathbf{i}',\sigma',m'})_n (u_{\mathbf{i},\sigma,m})_n\right]. \tag{2.51}$$

2.5 Experimental observables

We have laid out our general theory of multiorbital, two-dimensional superconductivity with impurity doping. Indeed, it is due to these impurities that interference patterns emerge, allowing us to visualise quantum matter at the atomic scale. Let us review the experimental observables of the theory, as probed via *scanning tunnelling microscopy* (STM).

2.5.1 Density of states

Even without impurities, the spectral function is always available to us. Experimentally, the spectrum is given by probing the differential conductance of the sample at a point. That is, electrons tunnel from the sample to the tip of the STM device, modulated by an applied bias voltage V

$$\frac{\mathrm{d}I}{\mathrm{d}V} \propto \mathrm{DOS}(eV;\mathbf{r}), \tag{2.52}$$

where the *density of states* at $\mathbf{r}_0$ is defined as

$$\mathrm{DOS}(\omega) \equiv -\frac{1}{\pi}\Im \mathcal{G}^R(\omega;\mathbf{r},\mathbf{r})\Big|_{\mathbf{r}=\mathbf{r}_0}, \tag{2.53}$$

where we define the shorthand for the trace of the Green's function as

$$\mathcal{G}^R(\omega;\mathbf{r},\mathbf{r}) \equiv \sum_{\sigma,m} \mathcal{G}_{\sigma,\sigma}^{R\,m,m}(\omega;\mathbf{r},\mathbf{r}). \tag{2.54}$$

2.5.2 Linecut

In order to facilitate the study of inhomogeneous samples, we would like to compare the spectrum at several points. Indeed, we can take the spectrum at different points and compare directly. Another possibility is the *linecut* which is the density of states, mapped as the stylus maps along a line e.g. radially outwards from an impurity.

2.5.3 Topography

In order to capture the *integrated density of states*, a *topography* may be performed, which consists of adjusting the tip height to maintain a constant current at constant bias voltage

$$\mathrm{IDOS}(\mathbf{r}, -eV) \equiv \int_{-eV}^{\omega_F} \mathrm{d}\omega \left[\Im\mathcal{G}^R(\omega; \mathbf{r}, \mathbf{r}) f(\omega) + \Im\bar{\mathcal{G}}^R(\omega; \mathbf{r}, \mathbf{r}) f(-\omega) \right]. \tag{2.55}$$

2.5.4 Density of states map

The *density of states map* (DOS map) consists of scanning the surface of the sample, obtaining a map of the DOS at each point. This is also known as the *local density of states*

$$\mathrm{LDOS}(\mathbf{r}, \omega) \equiv \lim_{\epsilon \to 0} \left[-\frac{1}{\pi} \Im\hat{\mathcal{G}}^R(\mathbf{r}, \mathbf{r}; \omega + i\epsilon) \right]. \tag{2.56}$$

2.5.5 Bogoliubov quasiparticle interference

The tunnelling conductance in momentum space $\mathrm{g}(\mathbf{q}, \omega)$ can also be probed. This is known as *Bogoliubov scattering interference amplitude* and is the Fourier transform of the local density of states

$$\mathrm{g}(\mathbf{q}, \omega) \equiv \mathrm{FT}\left(\mathrm{LDOS}(\omega)\right). \tag{2.57}$$

The intensity of this quantity is known as *Fourier-transform scanning tunnelling spectroscopy* (FT-STS) or *quasiparticle interference* and is defined by

$$\text{FT-STS}(\mathbf{q}, \omega) \equiv |\mathrm{g}(\mathbf{q}, \omega)|. \tag{2.58}$$

Impurities in a sample give rise to oscillations in the local density of states. Such *Friedel oscillations* have a characteristic wavelength of half the Fermi wavelength $\lambda_{\mathrm{Friedel}} = \frac{1}{2}\lambda_{\mathrm{Fermi}}$,

due to the scattering of the quasiparticles at the Fermi surface[3]. The quasiparticle interference intensity pattern therefore characterises the Fermi surface. Importantly, it is only a probe of the inner region of the Fermi surface, between $k_x, k_y \in \{-\pi/2, \pi/2\}$. Perhaps add figure explanation

The particle-hole symmetry of of interband scattering interference patterns depends on the relative sign of the energy gap on those bands. The energy (anti)symmetrised $\rho_\pm(\mathbf{q}, \omega)$ phase-resolved Bogoliubov scattering interference amplitudes [**?**] are defined as

$$\rho_\pm(\mathbf{q}, \omega) \equiv \Re\left[\mathrm{g}(\mathbf{q}, +\omega) \pm \mathrm{g}(\mathbf{q}, -\omega)\right], \tag{2.59}$$

which at the $\mathbf{q}$ for interband scattering, have distinct properties depending on the relative sign of the two gaps.

2.5.6 Scanned Josephson Tunnelling Microscopy

Cooper pair tunnelling is observable via *Josephson scanning tunnelling microscopy* (JSTM). In our model, it is defined as the *anomalous density of states*

$$\mathrm{ADOS}\left(\mathbf{r}, \omega\right) \equiv \lim_{\epsilon \to 0}\left[-\frac{1}{\pi}\Im\hat{\mathcal{F}}^R\left(\mathbf{r}, \mathbf{r}; \omega + i\epsilon\right)\right]. \tag{2.60}$$

2.5.7 Gap map

Analogously with the normal density of states, we can of course obtain results for the *Cooper pair map* or *gap map* from the *Cooper pair tunnelling* conductance.

2.5.8 Cooper pair interference

From the gap map, we can of course Fourier transform to the *Cooper pair interference* and obtain the momentum-dependence of the gap.

[3]Friedel oscillations are defined at zero applied bias ω. Pedantically, at nonzero bias the oscillations are technically not Friedel oscillations as we are probing quasiparticles away from the Fermi surface.

Chapter 3

Model of normal metal with impurities

In order to benchmark the general model with the known theory, we will first review the theory of Friedel oscillations in the normal metallic state. First we solve the problem of an electron at the Fermi surface, scattering off of an impurity. Our results are analytical, perturbative solutions in the continuum with infinite boundary conditions. This is then compared with our general model, applied to the particular. Indeed, we show qualitative agreement of the theories.

3.1 Analytical solution to an infinite, continuum model

3.1.1 Local density of states

The problem of an electron at the Fermi surface, scattering off of an impurity is solved using first-order perturbation theory. let us write the perturbation using the Dyson equation expansion of the retarded Green's function

$$\hat{\mathcal{G}}^R(\mathbf{r},\mathbf{r}';\omega) = \hat{\mathcal{G}}_0^R(\mathbf{r},\mathbf{r}';\omega) + \int \mathrm{d}\mathbf{r}''\hat{\mathcal{G}}_0^R(\mathbf{r},\mathbf{r}'';\omega)V(\mathbf{r}'')\hat{\mathcal{G}}^R(\mathbf{r}'',\mathbf{r}';\omega). \tag{3.1}$$

The Free Green's function is [Fiete and Heller]:

$$\hat{\mathcal{G}}_0^R(\mathbf{r},\mathbf{r}';\omega) = -i\frac{m^*}{2\hbar^2}H_0^{(1)}\left(k_\omega|\mathbf{r}-\mathbf{r}'|\right), \tag{3.2}$$

normal_state_band_structure.pdf

Figure 3.1: Energy dispersion of a two-dimensional metal described by a simple tight-binding model, such as Equation 2.1 with $s = \delta = 0$. The Fermi surface is projected onto the plane. Near the bottom of the band the Fermi surface is approximately quadratic. The Fermi surfaces corresponding to $\mu/t = -3.57$ and $\mu/t = -2.67$ are highlighted in white and pink, respectively. It is important to note that only the inner area, $k_x, k_y \in \{-\pi/2, \pi/2\}$ (thick dashed line) is observable via quasiparticle interference (QPI).

where $H_0^{(1)} = J_0 + iY_0$ is the Hankel function of the first-kind and k_ω is the momentum of these excitations. Near the bottom of the band, the dispersion relation Equation 3.25 becomes quadratic

$$\epsilon_0(k) = -2t\left[2 - \frac{k^2a^2}{2} + \mathcal{O}(k^4)\right] \tag{3.3}$$

$$= -4t + \frac{k^2}{2m^*} + \mathcal{O}(k^4) \tag{3.4}$$

where the effective mass $m^* \equiv 1/2ta^2$. This gives us an analytical expression for k_ω via the dispersion relation with applied bias ω

$$-\mu + \epsilon_0(k_\omega) = \omega \implies k_\omega a = \sqrt{\frac{\omega+\mu}{t} + 4}. \tag{3.5}$$

For concreteness, in the normal state $\Delta = 0$ at zero bias $\omega = 0$, with chemical potential $\mu = -3.88$, hopping parameter $t = 1$ and degenerate orbitals $s = \delta = 0$, the Fermi wavelength is

$$k_{\text{Fermi}}a \approx 0.346 \tag{3.6}$$

and correspondingly, the Friedel oscillations are described by

$$\lambda_{\text{Friedel}}/a = \frac{2\pi}{2k_{\text{Fermi}}a} \approx 9.07. \tag{3.7}$$

We plot the band structure for this situation in 3.1. In the absence of impurities, the local density of states is

$$\text{LDOS}(\mathbf{r},\omega) = -\frac{1}{\pi}\Im\hat{\mathcal{G}}_0^R(\mathbf{r},\mathbf{r};\omega) \tag{3.8}$$

$$= \frac{m^*}{2\pi} \equiv \rho_0 \tag{3.9}$$

since $J_0(x) \to 1$ as $x \to 0$. The first-order Green's function for the electron-impurity coupling V is given by solving the Dyson equation with a single recursion step. If we take the impurity to be located at the origin, the local density of states for small impurity coupling is given by

$$\text{LDOS}(\mathbf{r},\omega) = \rho_0 + \frac{(m^*)^2}{2\pi}VJ_0^{(1)}(k_\omega|\mathbf{r}|)\,Y_0^{(1)}(k_\omega|\mathbf{r}|)\,. \tag{3.10}$$

This analytical result from first order perturbation theory is shown in Figure **??** and the exact numerical results in Figures. **??**, **??**. We will now discuss our calculation for the exact numerical results.

3.1.2 Integrated local density of states

Carefully rederive for 2D or possibly remove this section. In the absence of impurities, the surface density of states is homogeneous. Surface impurities give rise to oscillations in the quasiparticle density, with a characteristic wavelength of half the Fermi wavelength $lambda_F/2 = \pi/k_F$, known as Friedel oscillations [**?**]. Friedel's eponymous waves may be derived by considering the surface of a material in its normal state around a point impurity. We derive a solution to the change in the non-interacting density of states $\delta\rho(\mathbf{x})$ due to the presence of an impurity, introduced in [**?**]. The first-order change to the density of states is

$$\delta\rho(\mathbf{x}) = iu_0 \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{i\omega 0^-} G(\mathbf{k},\omega) G(\mathbf{k}',\omega) \tag{3.11}$$

where u_0 is the interaction strength of the impurity, the summation is due to the momentum transformation of the real space scattering event, the inner integral is the frequency transformation of the real-time propagator, the exponential factor preserves causality and the Green's functions correspond to the ingoing and outgoing quasiparticles in the first-order expansion of the density of states. The inner integral gives rise to step-functions functions, or more physically, Fermi functions

$$\begin{aligned}
\delta\rho(\mathbf{x}) &= 2\pi u_0 \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}} \left[\frac{n_F(\mathbf{k})}{\zeta_{\mathbf{k}'} - \zeta_{\mathbf{k}}} + \frac{n_F(\mathbf{k}')}{\zeta_{\mathbf{k}} - \zeta_{\mathbf{k}'}} \right] && (3.12) \\
&= u_0 \Re \frac{(2\pi)^3}{V^2} \int_0^\infty k^2 \mathrm{d}k \int_0^\infty k'^2 \mathrm{d}k' \int_{-1}^{1} \mathrm{d}\cos\theta \int_{-1}^{1} \mathrm{d}\cos\theta' \ldots && (3.13) \\
&\quad \times \frac{n_F(k)}{k'^2 - k^2} e^{ikr\cos\theta} e^{-ik'r\cos\theta'}, && (3.14)
\end{aligned}$$

where for the purposes of carrying out the integral analytically, we have assumed the energy is quadratic, i.e. that we are near the bottom of the energy band. We then find

$$\begin{aligned}
\delta\rho(\mathbf{x}) &= u_0 \frac{(2\pi)^3}{V^2 r^2} \Re \int_0^\infty \mathrm{d}k \int_0^\infty \mathrm{d}k' \frac{kk'}{k'^2 - k^2} n_F(k) && (3.15)\\
&\quad \times \left[e^{ikr} - e^{-ikr}\right] \left[e^{-ik'r} - e^{ik'r}\right] && (3.16)\\
&= u_0 \frac{(2\pi)^3}{V^2 r^2} \Re \int_0^{k_F} \mathrm{d}k \int_0^\infty \mathrm{d}k' \frac{k}{2} \left[\frac{1}{k'-k} + \frac{1}{k'+k}\right] && (3.17)\\
&\quad \times \left[e^{ikr} - e^{-ikr}\right] \left[e^{-ik'r} - e^{ik'r}\right] && (3.18)\\
&= u_0 \frac{(2\pi)^4}{V^2 r^2} \Re \int_0^{k_F} \mathrm{d}k k \left[e^{ikr} - e^{-ikr}\right] \left[-ie^{-ikr}\right] && (3.19)\\
&= u_0 \frac{(2\pi)^4}{V^2} \left[-\frac{k_F \cos 2k_F r}{2r^3} + \frac{\sin 2k_F r}{4r^4}\right] && (3.20)\\
&= \sim u_0 \frac{\cos 2k_F r}{r^3}. && (3.21)
\end{aligned}$$

The result shows that if we have an impurity in the electron gas, then free electrons with Fermi wavelength λ_F will scatter. Peaks in the wavefunction correspond to two peaks in the charge density, therefore the observed electronic profile has a characteristic wavelength $\lambda_F/2$. Performing the integral in d dimensions, it can be shown that the envelope goes as $1/r^d$. We compare the integrated local density of states (ILDOS) with the local density of states (LDOS) in Figure **??**.

A quasiparticles near the bottom of the band takes on the Fermi energy $k_\omega \approx k_F$ and is the solution to the energy given by

$$-\mu + \epsilon_0(k_F) \approx 0 \implies k_F a = \sqrt{\frac{\mu}{t} + 4}. \tag{3.22}$$

3.2 Applying the general model to the particular: benchmarking the normal state

We have derived the very general phenomenon of Friedel oscillations analytically in the noninteracting normal state. We will now compare these with exact results from our general model. Satisfied that our results agree, we then study results for strongly-interacting regimes, where the free electron approximation no longer holds.

3.2.1 Impurities modulate the local density of states (Friedel oscillations)

3.2.2 Density functional theory of impurities

3.2.3 Fourier transformation of the mean-field Hamiltonian

Possibly omit unnecessary section Let us explicitly relate the $\mathbf{r}$-space picture to the $\mathbf{k}$-space via a discrete Fourier transform

$$\hat{c}^\dagger_{\mathbf{k}m\sigma} = \frac{1}{\sqrt{N}} \sum_{\mathbf{i}\in\Omega} e^{+i\mathbf{k}\cdot\mathbf{r_i}} \hat{c}^\dagger_{\mathbf{i}m\sigma}, \tag{3.23}$$

where $\mathbf{k} \in \mathcal{K} \equiv \left\{ \left(\frac{2\pi p}{N_x}, \frac{2\pi q}{N_y} \right) \middle| 0 \leqslant p < N_x,\, 0 \leqslant q < N_y \right\}$. The position vector $\mathbf{r_i}$ is defined in terms of lattice spacings a_x, a_y as $\mathbf{r}_i = (a_x i_x, a_y i_y)$. The Hamiltonian in $\mathbf{k}$-space is then

$$\begin{aligned}
\mathcal{H} = \sum_{\mathbf{k}\in\mathcal{K}} \sum_{m\in\{+,-\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} & \left[-\mu_m \hat{c}^\dagger_{\mathbf{k}m\sigma} \hat{c}_{\mathbf{k}m\sigma} + \delta \hat{c}^\dagger_{\mathbf{k}m\sigma} \hat{c}_{\mathbf{k}\bar{m}\sigma} \right] \\
& + \sum_{\mathbf{k},\mathbf{q}\in\mathcal{K}} \sum_{m\in\{+,-\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[\tilde{V}(\mathbf{k}-\mathbf{q}) \hat{c}^\dagger_{\mathbf{k}m\sigma} \hat{c}_{\mathbf{q}m\sigma} \right] \\
& + \sum_{\mathbf{k}\in\mathcal{K}} \sum_{m,n\in\{+,-\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[\epsilon_0(\mathbf{k}) \hat{c}^\dagger_{\mathbf{k}m\sigma} \hat{c}_{\mathbf{k}n\sigma} \right] \\
& + \sum_{\mathbf{k}\in\mathcal{K}} \sum_{m,n\in\{+,-\}} \sum_{\sigma,\tau\in\{\uparrow,\downarrow\}} \left[\Delta^{mn}_{\mathbf{k}\sigma\tau} \hat{c}^\dagger_{\mathbf{k}m\sigma} \hat{c}^\dagger_{-\mathbf{k}n\tau} + \text{h.c.} \right]
\end{aligned} \tag{3.24}$$

where

$$\epsilon_0(\mathbf{k}) = -2t_{m,n} \left(\cos k_x a_x + \cos k_y a_y \right). \tag{3.25}$$

and

$$\tilde{V}(\mathbf{k}-\mathbf{q}) = V \frac{1}{N} \sum_{i'\in\Omega'} e^{-i(\mathbf{k}-\mathbf{q})\mathbf{r}_{i'}} \tag{3.26}$$

3.2.4 Quasiparticle interference

benchmarking_ns.pdf

Figure 3.2: Top: The local density of states $\mathrm{LDOS}(\mathbf{r}, \omega + i\epsilon)$ at r_n at zero bias ($\omega = 0$) with convergence factor $\epsilon = 0.1$ as a function of square systems of various sizes n with periodic boundary conditions. The system is the normal state ($\Delta = 0$), single band ($s = \delta = 0$) with chemical potential $\mu = -3.57$ and normalised by the hopping parameter ($t = 1.0$). The local density of states are homogeneous as there are no impurities. We observe the finite-size effects attenuate with increasing system size n. // Bottom: Memory stored in the matrix representation of the Hamiltonian and execution time of the simulation.

LDOS_line_mosaic_normal_state_varying_n.pdf

Figure 3.3: The local density of states $\mathrm{LDOS}(\mathbf{r}, \omega + i\epsilon)$ along a line from $r_0 = [0, 0]$ to $r_{\mathrm{diag}} = [(n-1)/2, 0]$ and $r_{\mathrm{horiz}} = [(n-1)/2, (n-1)/2]$ at zero bias ($\omega = 0$) with convergence factor $\epsilon = 0.1$ for square systems of various sizes, as labelled in the legend, with periodic boundary conditions. The system is the normal state ($\Delta = 0$), single band ($s = \delta = 0$) with chemical potential $\mu = -3.57$, normalised by the hopping parameter ($t = 1.0$) and with an impurity at the centre of the lattice r_0 with coupling strength $V = 1.21$. The local density of states is plotted along a straight path from the centre of the lattice to half-way to the nearest neighbour, as a function of the distance in sites (top left) and as a function of the distance in sites normalised by the system size (top right); and similarly to half-way to the next-nearest neighbour (bottom left and right). We observe Friedel oscillations, that is to say, a modulation in the local density of states due to the translational symmetry breaking.

LDOS_line_Analytical_versus_numerical.pdf

Figure 3.4: The local density of states at bias $\omega = 0$ and resolution $\epsilon = 0.1$ along a line from $[-44, 0]$ to $[44, 0]$ with an impurity located at $r_0 = [0, 0]$ with coupling strength $V/t = 1.21$ at various chemical potentials μ. The model is spinless and single band, hence superconductivity does not form. The numerical model consists of 79 sites along each dimension with periodic boundary conditions. The analytical result is the first-order expansion of the continuum Dyson equation with infinite boundary conditions (Equation 3.10) of a free-electron with an effective mass, an approximation which holds near the bottom of the band $\mu/t \gtrapprox -4$. The constant zeroth order term deviates from the exact numerical result as we move higher up the band, whilst the first order term reproduces the Friedel oscillations quantitatively. The non-self-consistent numerical result of the density is a stationary solution, as its degeneracy with the self-consistent solutions shows. The continuum result depends on a renormalisation of the potential which has not be taken into account and the free electron approximation.

LDOS_r_vs_V.pdf

Figure 3.5: The local density of states far away from the impurity does not depend strongly on the impurity coupling strength V/t. The model consists of 43×43 sites and chemical potential $\mu/t = -3.57$.

QPI_Fermi_surface_normal_state.pdf

Figure 3.6: Quasiparticle interference (QPI) of the spinless, bandless normal state with $n = 79$ sites at zero applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.1$ The impurity, necessary for quasiparticle interference, is placed at $r = (0, 0)$ with coupling strength $V/t = 1.21$. The QPI interference pattern at zero bias probes the inner area $\{-\pi/2, \pi/2\}$ of the Fermi surface.

QPI_Fermi_surface_normal_state_multiorbital.pdf

Figure 3.7: Quasiparticle interference (QPI) of a spinless, two-band normal state with $n = 57$ sites (maximum size on 64Gb RAM) at zero applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.1$. The bands have a rigid shift $s/t = 0.05$ and hybridisation $\delta/t = 0.075$. The impurity, necessary for quasiparticle interference, is placed at $r = (0, 0)$ with coupling strength $V/t = 1.21$. The QPI interference pattern at zero bias probes the inner area $\{-\pi/2, \pi/2\}$ of the Fermi surface.

LDOS_line_normal_state_varying_mu.pdf

Figure 3.8: The frequency of the Friedel oscillations in the local density of states increase as we move higher up the band with increasing μ/t. The number of sites along each dimension and the impurity coupling are fixed at 43 and $V/t = 1.21$, where we have normalised by the hopping. The bias and resolution are $\omega = 0$ and $\epsilon = 0.1$, respectively.

Spectrum_3D_normal_state_varying_V.pdf

Figure 3.9: Spectrum with resolution $\epsilon = 0.1$ of the normal state to study the effect of an impurity located at the centre as a function of its coupling strength V/t. The model consists of 43×43 sites and chemical potential $\mu/t = -2.67$.

LDOS_mosaic_normal_state.pdf

Figure 3.10: The local density of states in the normal state with chemical potential and coupling strengths $\mu/t = -3.57$ and $V/t = 1.21$, with 43 sites along each axis. Clockwise from top left: single central impurity; two impurities $(-3, 3)$ and $(3, -3)$; random arrangement of 20 impurities; and a *corral* of 15 impurities arranged in a circle with radius 14.5 sites. Top left figure, the arrow denote the two points probed in Figure 3.11.

Spectrum_Normal_state_43_43_-2.67_0_0_1.00_0_1.21e+00.pdf

Figure 3.11: The spectrum in the normal state, taken at the two points given by the arrows in Figure 3.10, top left frame: the impurity (navy) and five atoms away (green). The impurity potential and chemical potential are set to $V/t = 1.21$ and $\mu/t = -3.57$. In the next Figure 3.12 we study a linecut between these two points.

Linecut_Normal_state_43_43_-2.67_0_0_1.00_0_1.21e+00.pdf

Figure 3.12: *The temples of Tamil Nadu.* A *linecut* is the local density of states at several equally spaced intervals along a straight line, moving away from the impurity. We see clearly the van Hove singularity as well as a peak due to the impurity. The impurity potential and chemical potential are set to $V/t = 1.21$ and $\mu/t = -3.57$. The line is shown in the top left frame of Figure 3.10, where the impurity (navy) and five atoms away (green) are given by the arrows.

LDOS_3D_Normal_state_corral_43_43_-3.57_0_0_1.00_0_1.21e+00.pdf

Figure 3.13: The local density of states of the corral depicted three-dimensionally in order to clearly expose the standing waves in the electron density due to the ring of 15 impurities. The chemical potential is $\mu/t = -3.57$ and the coupling strength is $V/t = 1.21$, with 43 sites along each axis. The bias energy is $\omega/t = 0$ with resolution $\epsilon/t = 0.1$

Self-consistent_singlet_n=5_T=varying.pdf

Figure 3.14: Self-consistent results for one band without impurities, chemical potential $\mu/t = -2.67$ and 5 sites along each axis. The critical temperature is in good agreement with the BCS conventional theory $T_c^{\mathrm{BCS}} = \Delta_0/1.76$. The ϕ-fields are degenerate and comparison less temperature dependant than the pairing potential Δ, as expected in conventional BCS theory. The absolute convergence factor is $\varepsilon = 10^{-3}$.

Self-consistent_singlet_n=25_T=varying.pdf

Figure 3.15: Results for $n = 25$.

Self-consistent_singlet_n=51_T=varying.pdf

Figure 3.16: Results for $n = 51$ with $\varepsilon = 10^{-2}$ order of magnitude less than in the $n = 5, 25$ simulations.

Self-consistent_singlet_n=5_mu=varying.pdf

Figure 3.17: Self-consistent results for one band without impurities at zero temperature $T = 0$, with 5 sites along each axis. The ϕ-fields are degenerate. The absolute convergence factor is $\varepsilon = 10^{-3}$.

Self-consistent_singlet_n=25_mu=varying.pdf

Figure 3.18: Results for $n = 25$

Self-consistent_singlet_n=51_mu=varying.pdf

Figure 3.19: Results for $n = 51$ with $\varepsilon = 10^{-2}$ order of magnitude less than in the $n = 5, 25$ simulations.

Self-consistent_singlet_n_varying.pdf

Figure 3.20: Self-consistent results for one band without impurities at zero temperature $T = 0$. The chemical potential is $\mu/t = -2.67$. The ϕ-fields are degenerate. The absolute convergence factor is $\varepsilon = 10^{-3}$.

Self-consistent_singlet_single_impurity.pdf

Figure 3.21: Self-consistent results for one band with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$.

Self-consistent_multiorbital_singlet_single_impurity.pdf

Figure 3.22: Self-consistent results for a multiorbital model with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$.

Self-consistent_triplet_single_impurity.pdf

Figure 3.23: Self-consistent results for a non-unitary triplet model with no impurity (blue, crosses), a single impurity (green, pluses) and a magnetic impurity (red, dots) with 43 sites along each axis, both with coupling strength $V/t = 1.2$, all at zero temperature $T = 0$ and chemical potential $\mu/t = -2.67$. The Hubbard $U/t = 2.8$, the self-consistent absolute convergence factor is $\varepsilon = 10^{-3}$.

LDOS_FT_mosaic_normal_state_random_impurities_mu_-3.57.pdf

Figure 3.24: Quasiparticle interference (magnitude of the Fourier transform) of the normal state local density of states. Top left: Fourier transform of Figure 3.10, a single central impurity. Top right: 20 random impurities. Bottom: averaging over 8 and 100 simulations of 20 different random impurities. The chemical potential is $\mu/t = -3.57$ and the impurity coupling strength is $V/t = 1.21$. This local electron density is a function of the momentum of the Friedel waves and is 'double' the Fermi surface in the following sense $\boldsymbol{k}_{Fermi} = \frac{1}{2}\boldsymbol{k}_{Friedel}$. In the experimental literature, quasiparticle interference is performed by averaging over many trial experiments to remove noise from the image, as we simulate here. In our theoretical simulations however, we can use a single, central impurity to produce noiseless quasiparticle interference patterns. Such results differ only in magnitude, as we have shown here by using two different colourbars, one for the single impurity and another for the three examples of many-impurity averaging.

Chapter 4

Models of multi-orbital superconductivity

4.1 Model of multi-orbital spin-singlet superconductivity

Spin-singlet pairing is defined in terms of the pairing tensor

$$\mathbf{\Delta} = \Delta_{\mathbf{i}} \, (i\sigma_y \otimes \mathbb{1}), \tag{4.1}$$

where the first sector in the tensor product represents the spin channels $\sigma = \uparrow, \downarrow$ and the second sector represents the orbital channels $m \in \{+, -\}$. $\Delta_{\mathbf{i}}$ is an on-site pairing amplitude, which varies in space only in the presence of impurities.

4.1.1 Model and reproducing known results

4.2 Model of multi-orbital (non-)unitary spin-triplet superconductor with impurities

Chapter 5

Su-Schrieffer-Heeger model with multiorbital superconductivity

As a first departure into the search for systems exhibiting topological superconductivity, we begin with a topologically non-trivial system. A canonical topological system is the Su-Schrieffer-Heeger (SSH) model of a spinless, one-dimensional dimer lattice of fermions with staggered hopping. The model obeys a *chiral symmetry*, exhibiting two topological phases. In the topologically non-trivial phase, a pair of edge states emerges at the boundaries, characterised by a *winding number*. We will consider the effect of a spin-triplet, multiorbital attractive interaction within the unit cell, and a Coulomb repulsion to nearest-neighbours outside of the unit cell.

5.1 Su-Schrieffer-Heeger model with spin

The Su-Schrieffer-Heeger model with additional spin degrees of freedom is given by the following Hamiltonian

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \sum_{m\in\{A,B\}} \left[-\mu \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m}\right] + \sum_{\sigma\in\{\uparrow\downarrow\}} \left[V \hat{c}^\dagger_{0,\sigma,m} \hat{c}_{0,\sigma,m}\right] + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow\downarrow\}} \left[-v \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} - w \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} + \text{h.c.}\right], \quad (5.1)$$

where μ is the chemical potential, v is the intracell, interorbital hopping parameter, w is the intercell, interorbital hopping and V is an impurity coupling strength. The model boundary conditions are periodic, and therefore, in the topological state, we expect edge modes to arise around the impurity. In the trivial state, with $v > w$, the unit cell has the strucure displayed in Figure 5.1.

We investigate the phases of the model in Figure 5.2. Importantly, we may have naivly expected the addition of a spin quantum number to the SSH to be trivial i.e. simply doubling the degrees of freedom of the original SSH model. Performing the calculation, however, reveals nontrivial magnetic effects at $\mu = 0$.

5.2 Non-unitary spin-triplet Su-Schrieffer-Heeger model

The non-unitary spin-triplet extension of the Su-Schrieffer-Heeger model is given by

$$\hat{\mathcal{H}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-U_v \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} + U_w \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \right], \tag{5.2}$$

where U_v is the spin-triplet interaction and U_w is a Coulomb repulsion.

The mean-field solution is

$$\begin{aligned} \hat{\mathcal{H}}^{\text{mf}}_{\text{SSH-triplet}} = \hat{\mathcal{H}}_{\text{SSH}} + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \Bigg[& \sum_{m\in\{A,B\}} \left[\phi_{x,\sigma,m} \hat{c}^\dagger_{x,\sigma,m} \hat{c}_{x,\sigma,m} \right] \\ & + \Phi^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}_{x,\sigma,B} + \Phi^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}_{x+1,\sigma,A} + \Delta^v_{x,\sigma} \hat{c}^\dagger_{x,\sigma,A} \hat{c}^\dagger_{x,\sigma,B} + \Delta^{v\dagger}_{x,\sigma} \hat{c}_{x,\sigma,B} \hat{c}_{x,\sigma,A} \\ & + \Delta^w_{x,\sigma} \hat{c}^\dagger_{x,\sigma,B} \hat{c}^\dagger_{x+1,\sigma,A} + \Delta^{w\dagger}_{x,\sigma} \hat{c}_{x+1,\sigma,A} \hat{c}_{x,\sigma,B} \Bigg], \end{aligned} \tag{5.3}$$

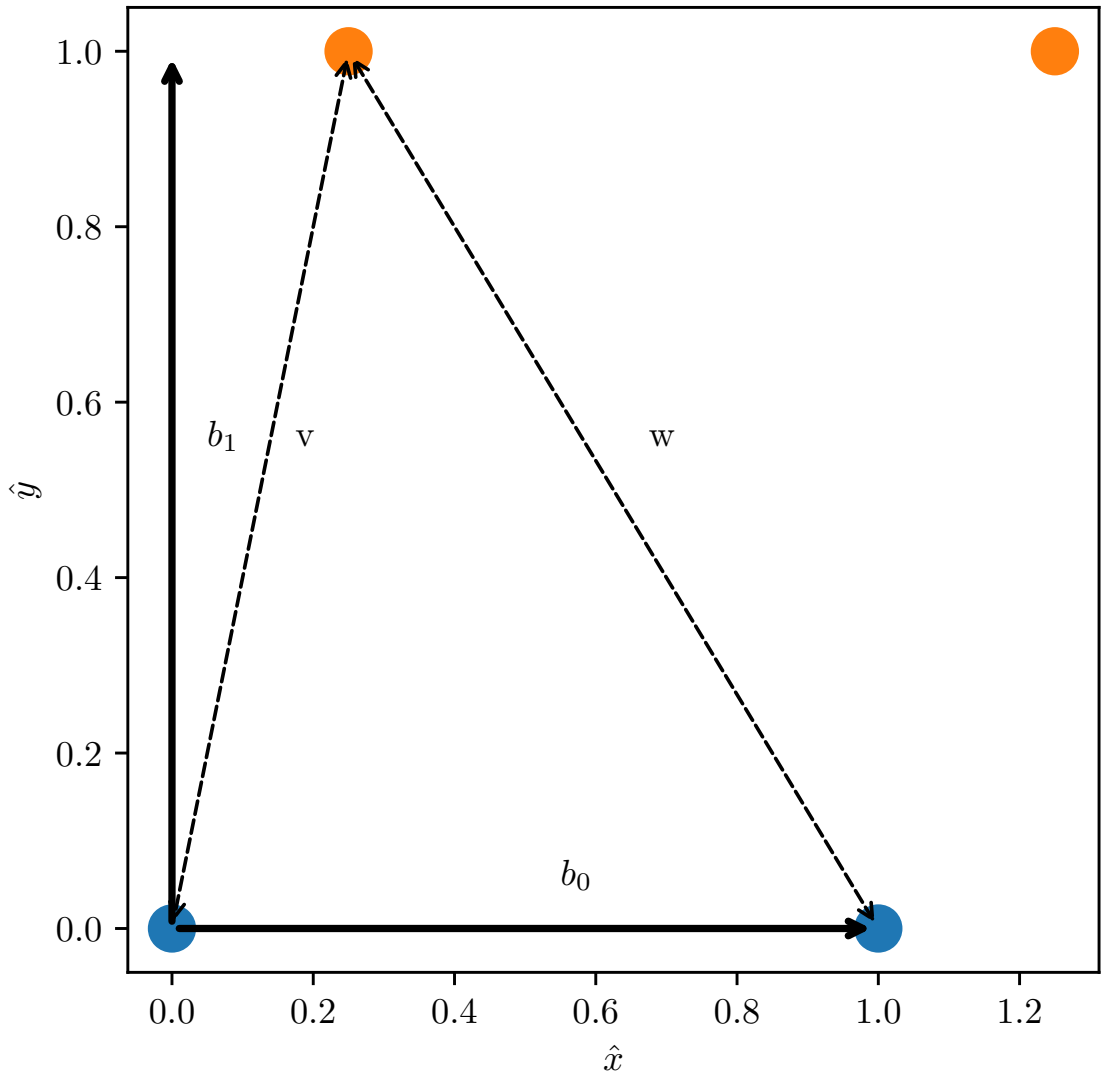

Figure 5.1: The dimer unit cell of the SSH model is depicted, together with hopping parameters w and v, as well as the lattice basis b_0 and b_1. Atoms A and B are colored in blue and orange, respectively.

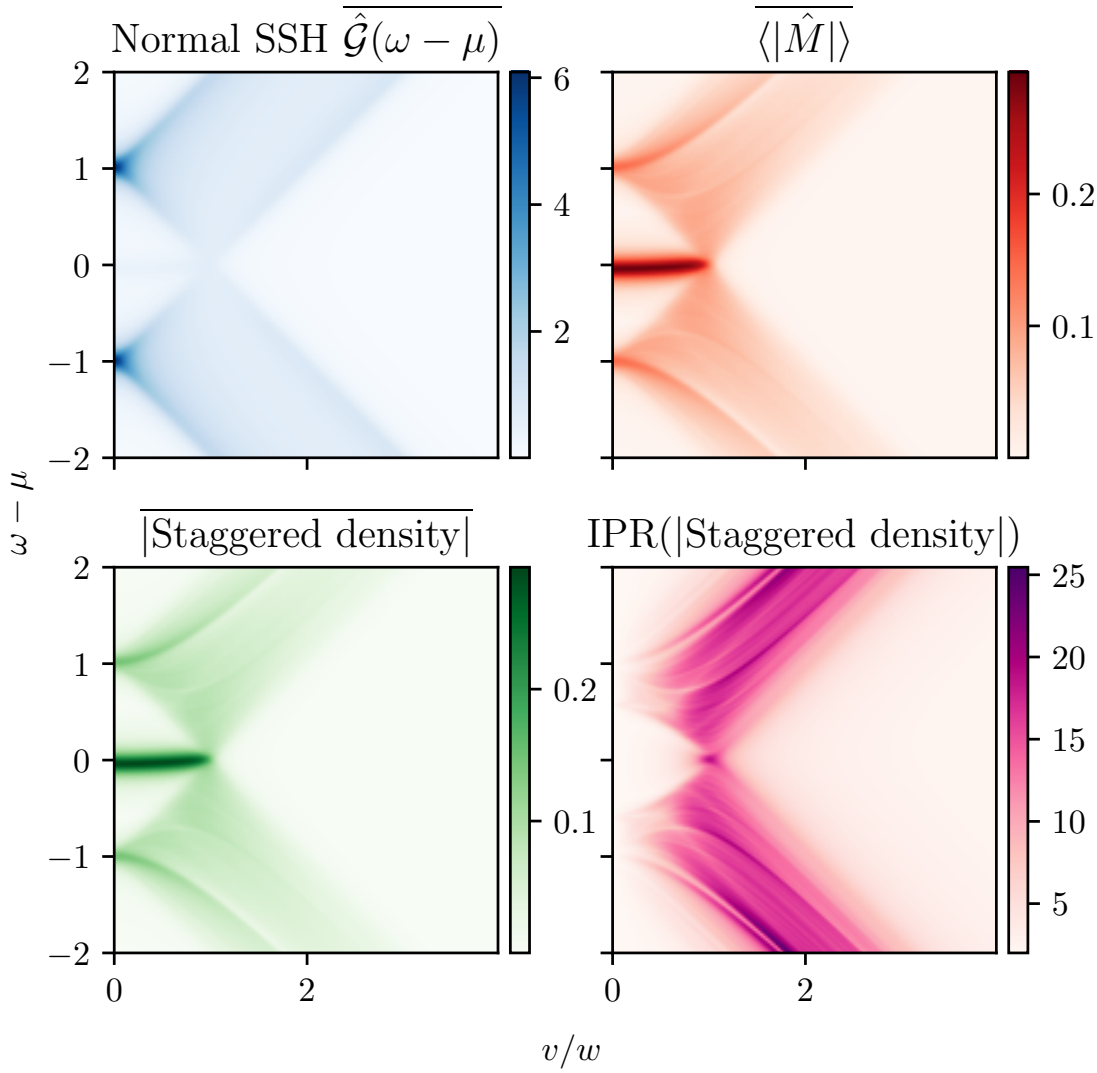

Figure 5.2: Phase diagram of the SSH model with $n = 43$ dimers and periodic boundary conditions. The density of states are plotted as a function of the applied bias energy $\omega - \mu$, noting that the chemical potential contributes only an overall shift, and as a function of the intracell, interorbital hopping parameter v/w. The topological transition occurs at $v/w = 1$ and notice the faint density at $\omega = 0$ below the transition point, due to the edge states. A large impurity coupling $V = 20.28$ gaps out any modes on the middle site, effectively cutting the closed chain into open boundary conditions. In order to have a measure of the topological modes, we plot the mean absolute magnetism and the mean absolute staggered density, which signal charge density waves. The inverse participation of the staggered density is a measure of long-range order, demonstrating that the topological modes are localised around the impurity.

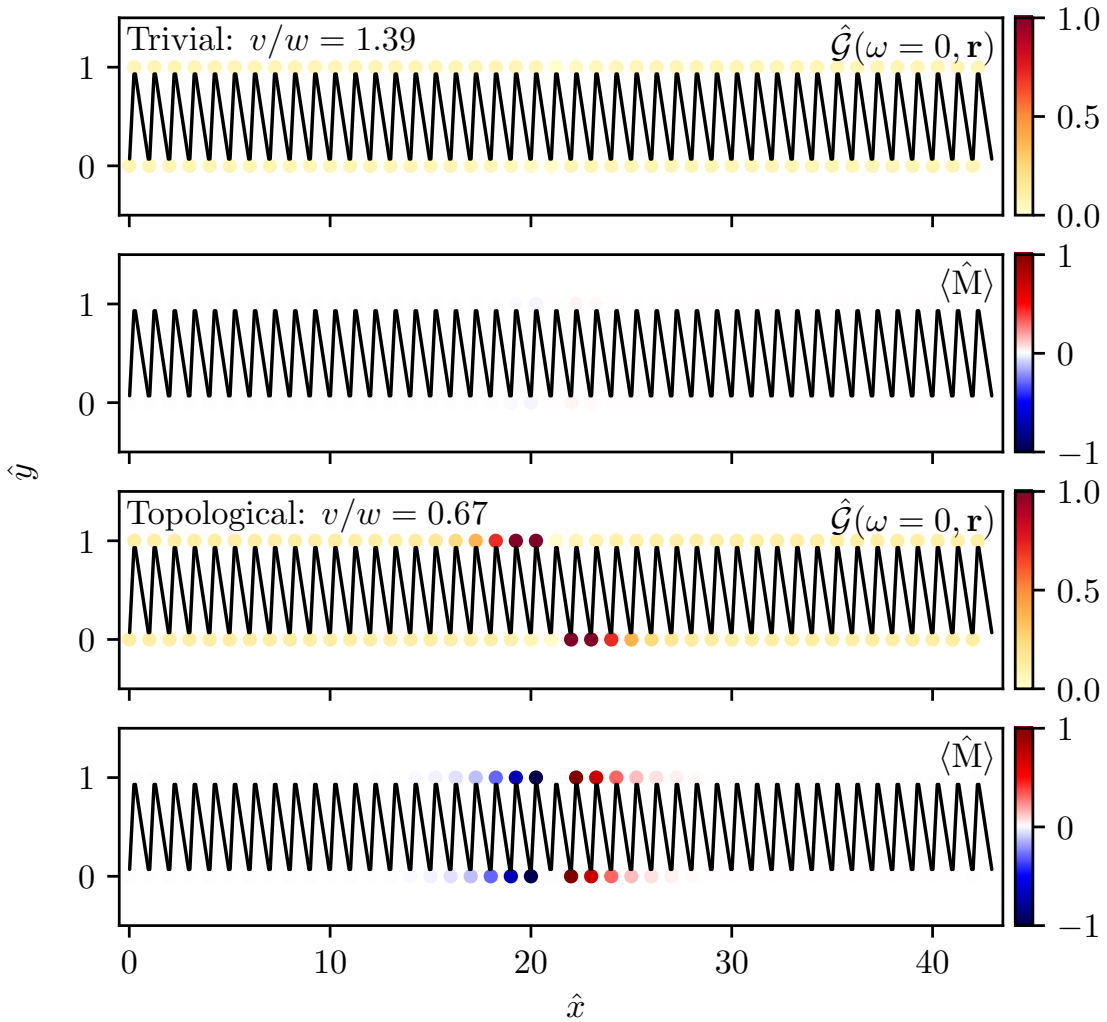

Figure 5.3: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = 0$ and strong impurity coupling strength $V = 20.28$ on the middle site, in order to cut the chain without weak coupling effects. The dark lines represent hopping bonds and the sites are denoted by circles with color corresponding to density. Importantly, we note non-trivial orbital and magnetic characteristics to the edge modes.

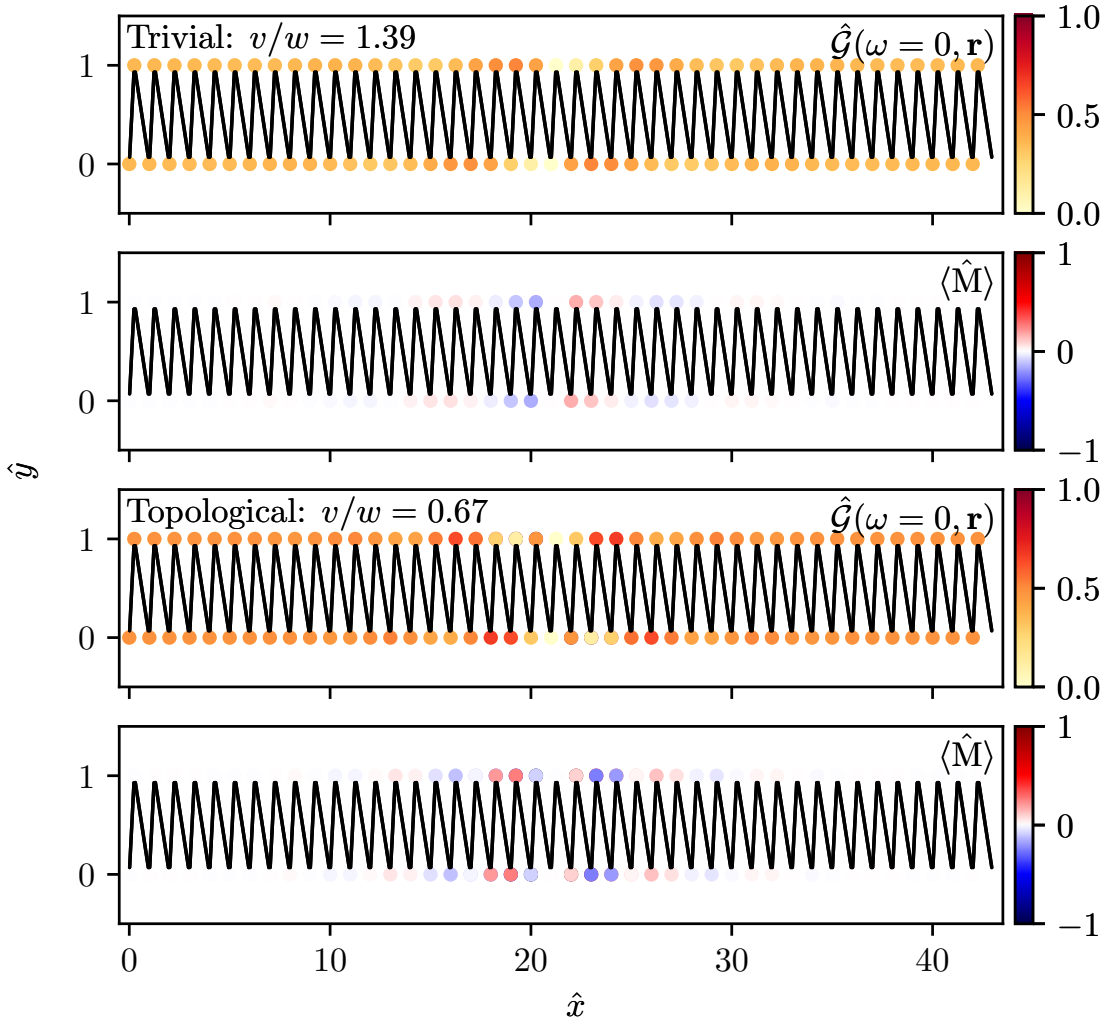

Figure 5.4: The local density of states and magnetism of the SSH model in the normal state with chemical potential $\mu = -0.57$ and strong impurity coupling strength $V = 20.28$ on the middle site. Away from zero chemical potential, we observe a charge density wave in the topological phase.

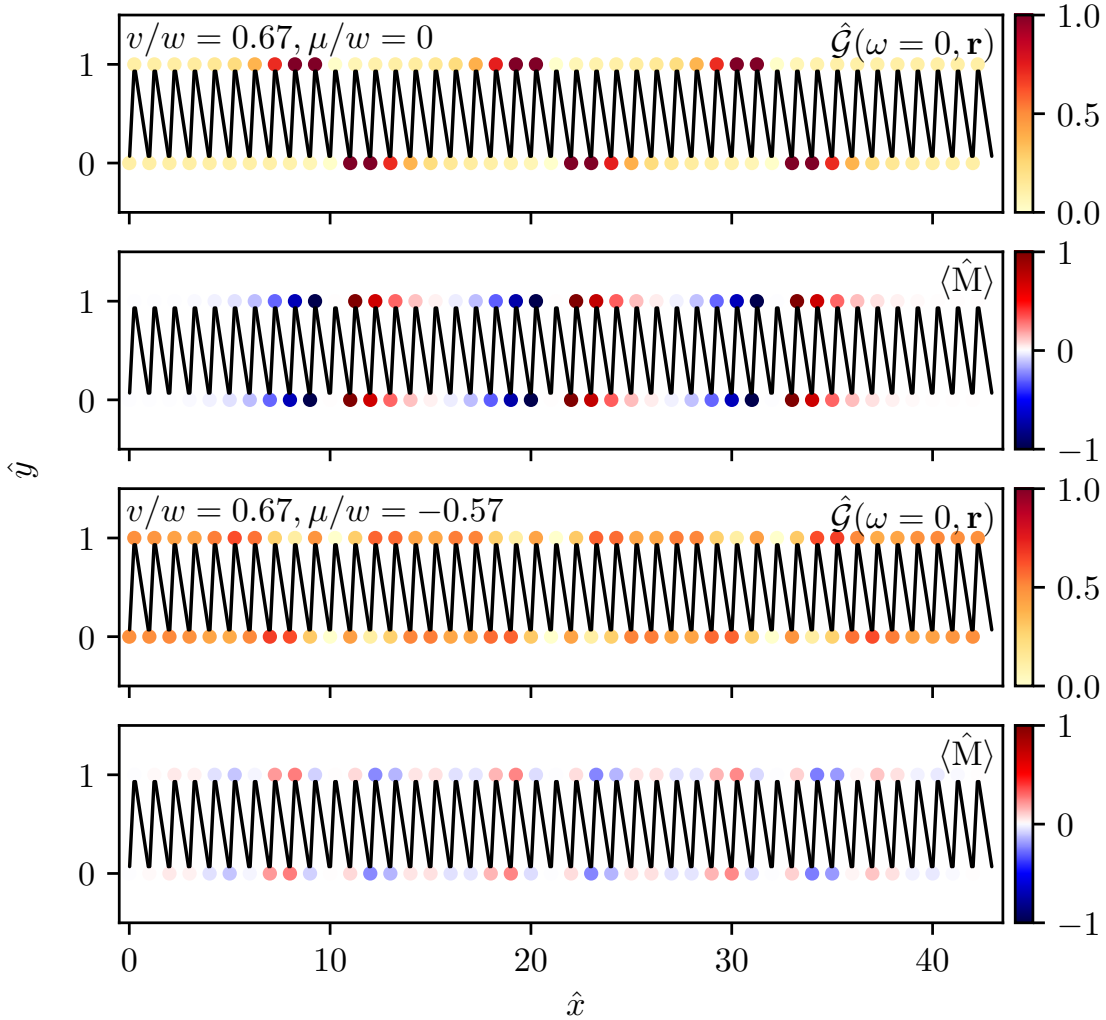

Figure 5.5: The local density of states and magnetism of the SSH model in the normal state with two impurites with strong coupling strengths $V = 20.28$ at $x = \pm 11$. Curiously, a kink forms in the middle $x = 0$, where there is no impurity.

where the Hartree, Fock and Gorkov fields are defined as

$$\phi_{x,\sigma,m} = (U_w - U_v)\hat{c}^{\dagger}_{x,\sigma,m}\hat{c}_{x,\sigma,m} \tag{5.4}$$

$$\Phi^{v}_{x,\sigma} = -U_v\hat{c}^{\dagger}_{x,\sigma,A}\hat{c}_{x,\sigma,B} \tag{5.5}$$

$$\Phi^{w}_{x,\sigma} = \hat{U}_w c^{\dagger}_{x,\sigma,B}\hat{c}_{x+1,\sigma,A} \tag{5.6}$$

$$\Delta^{v}_{x,\sigma} = -U_v\hat{c}_{x,\sigma,B}\hat{c}_{x,\sigma,A} \tag{5.7}$$

$$\Delta^{v\dagger}_{x,\sigma} = U_w\hat{c}^{\dagger}_{x,\sigma,A}\hat{c}^{\dagger}_{x,\sigma,B} \tag{5.8}$$

$$\Delta^{v}_{x,\sigma} = -U_v\hat{c}_{x+1,\sigma,A}\hat{c}_{x,\sigma,B} \tag{5.9}$$

$$\Delta^{w\dagger}_{x,\sigma} = U_w\hat{c}^{\dagger}_{x,\sigma,B}\hat{c}^{\dagger}_{x+1,\sigma,A} \tag{5.10}$$

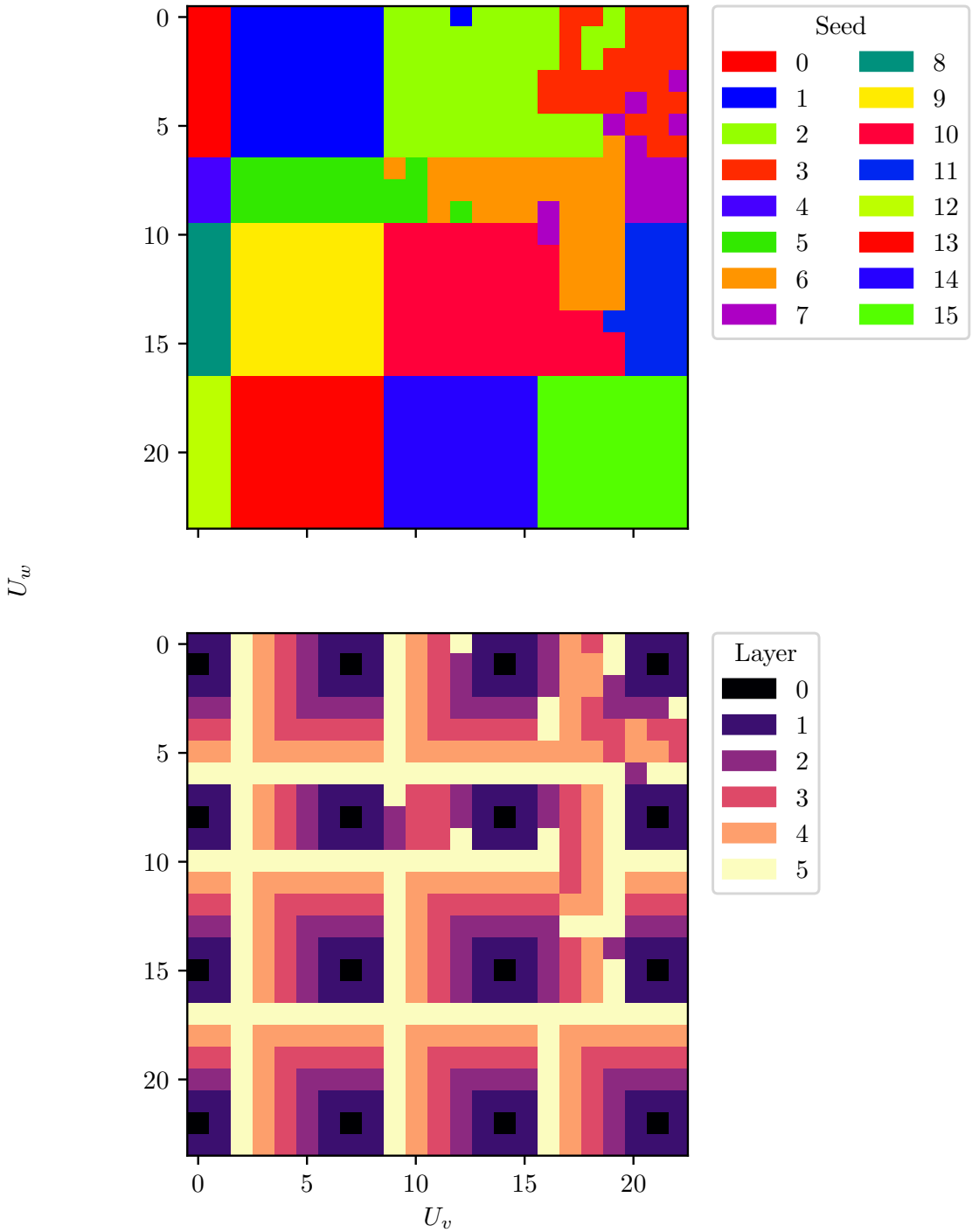

Figure 5.6: Building the phase diagram for a spin-rtiplet SSH model. Above: seeds are perculated evenly spaced throughout phase space. At each location, two seeds are placed, corresponding to the unitary and non-unitary initial mean-fields. Below: from these seeds (refered to as Layer 0), layers are grown outwards from their parent seed. The outer layers of various seeds overlap, and that with the minimum free energy is selected.

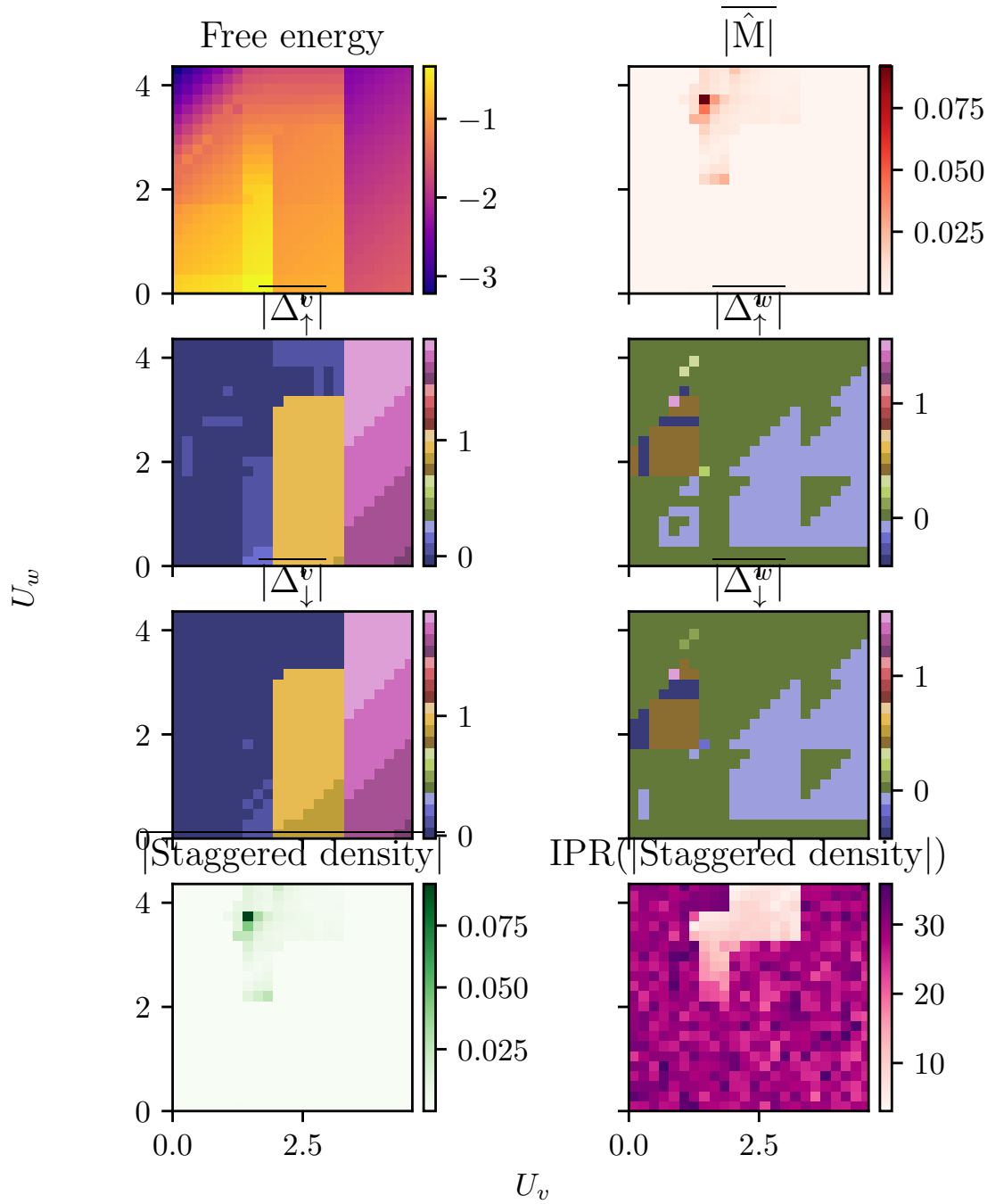

Figure 5.7: The phase diagrams for an SSH model with multiorbital, equal-spin superconductivity interaction and no impurity. Two initial pairing fields are given to the self-consistent loop: non-unitary triplet and equal-spin triplet. The solution with the lowest free energy is selected. We observe that the mean-field spin-sectors are equal $\overline{\Delta^{\uparrow\uparrow}_{v/w}} = \overline{\Delta^{\downarrow\downarrow}_{v/w}}$, hence the equal-spin triplet theory is favoured. We also observe spontaneous formation of charge density waves, localised in space.

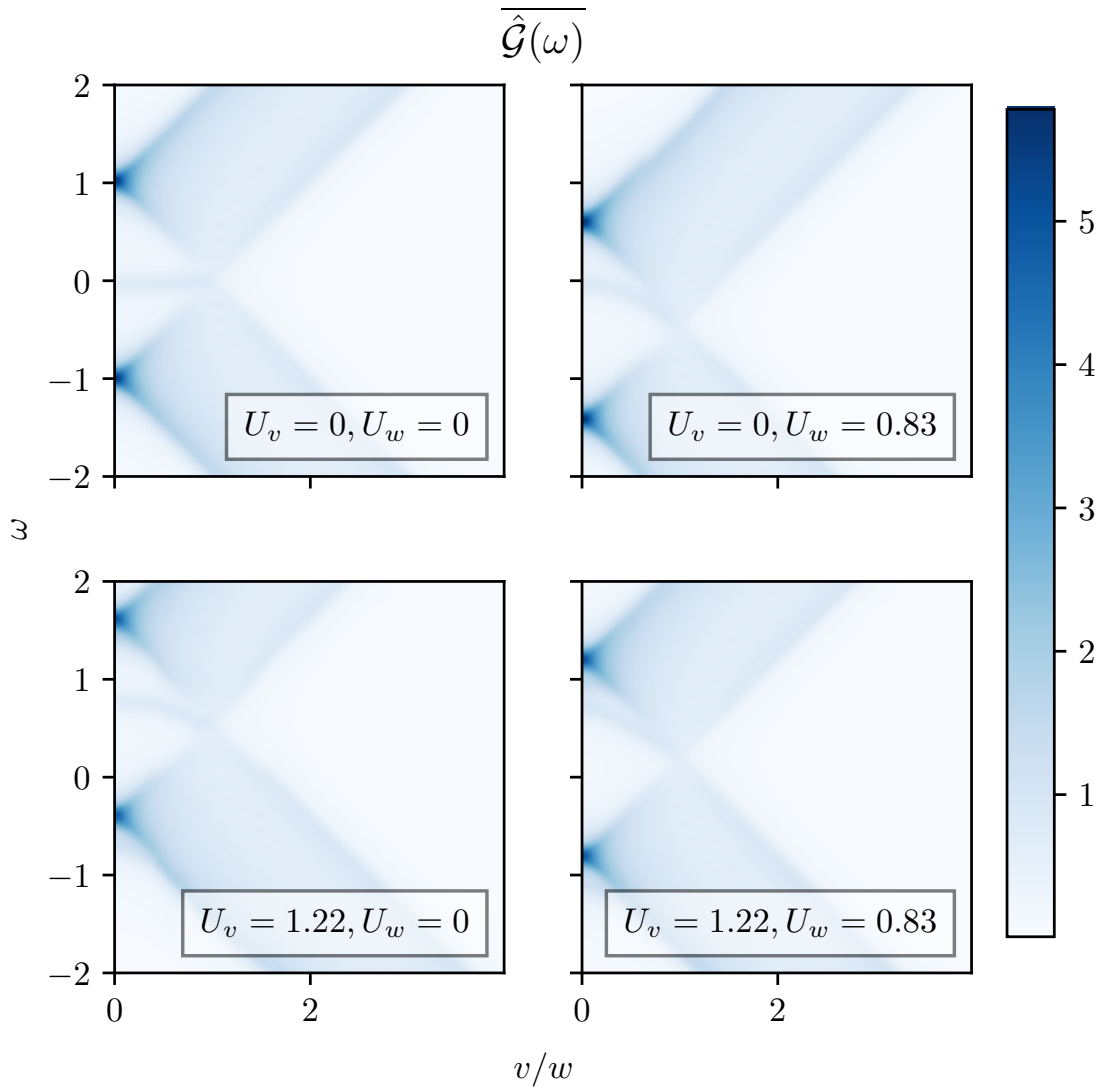

Figure 5.8: Mean value of the local density of states. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

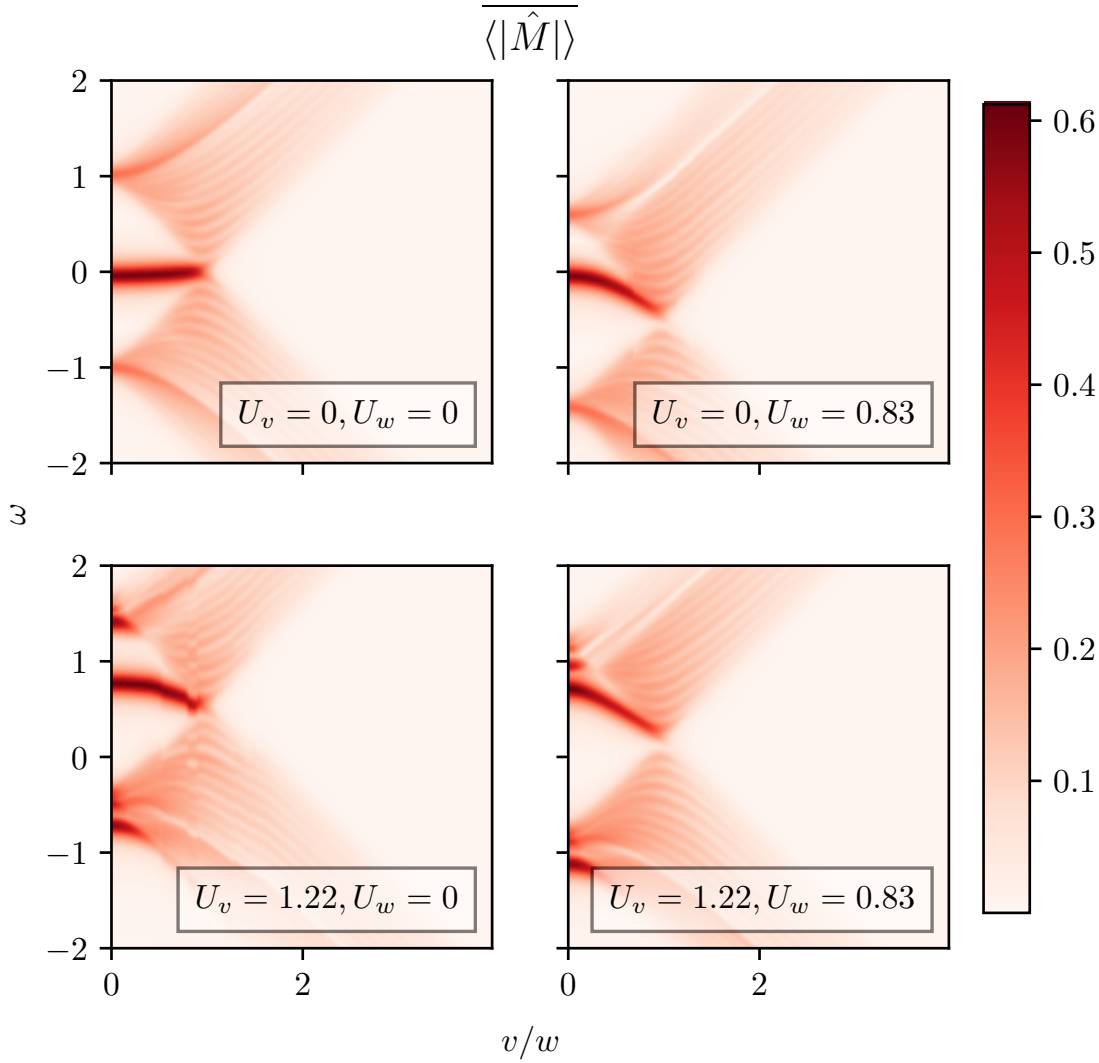

Figure 5.9: Mean absolute value of magnetism. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

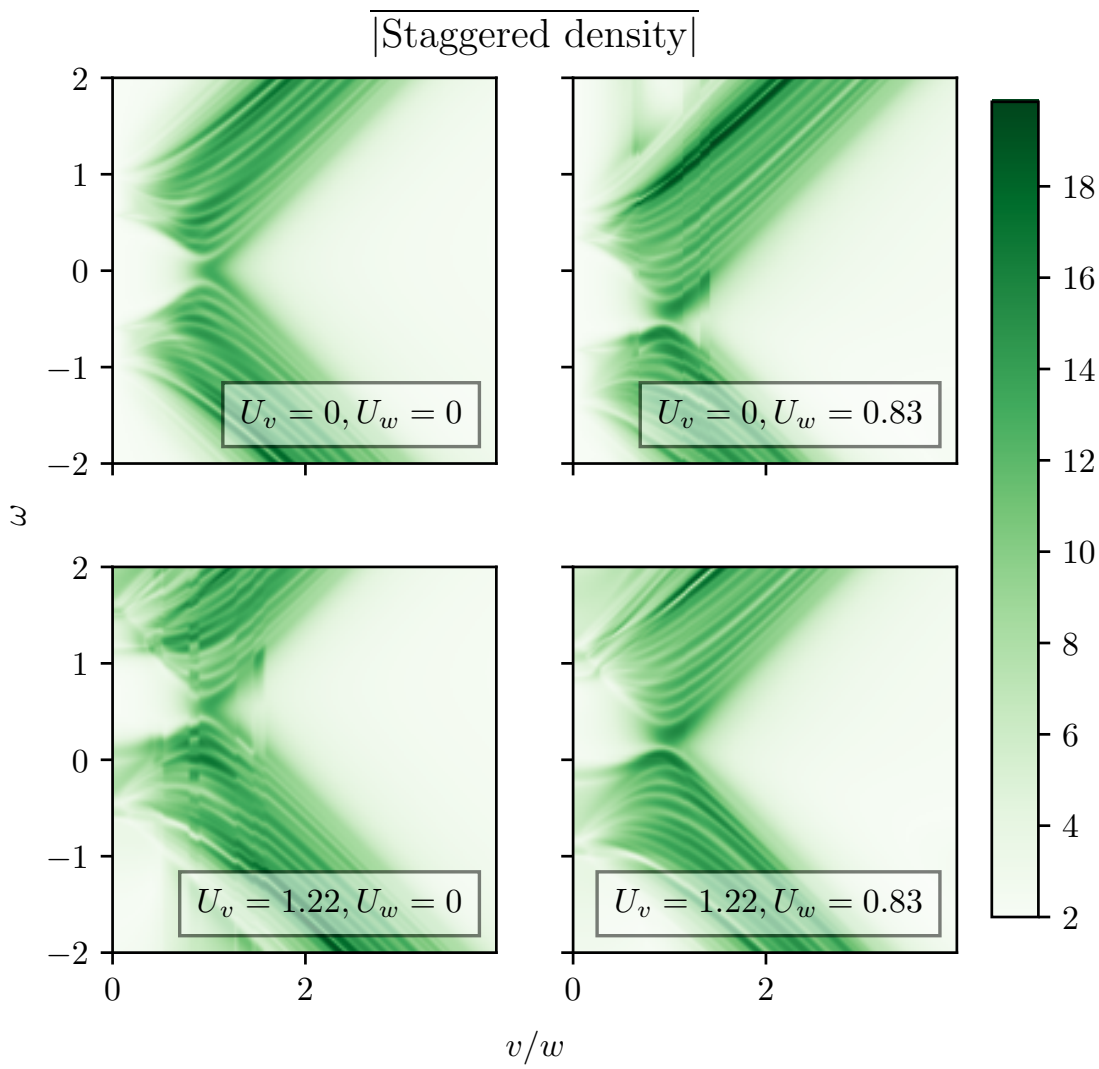

Figure 5.10: Mean staggered density. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

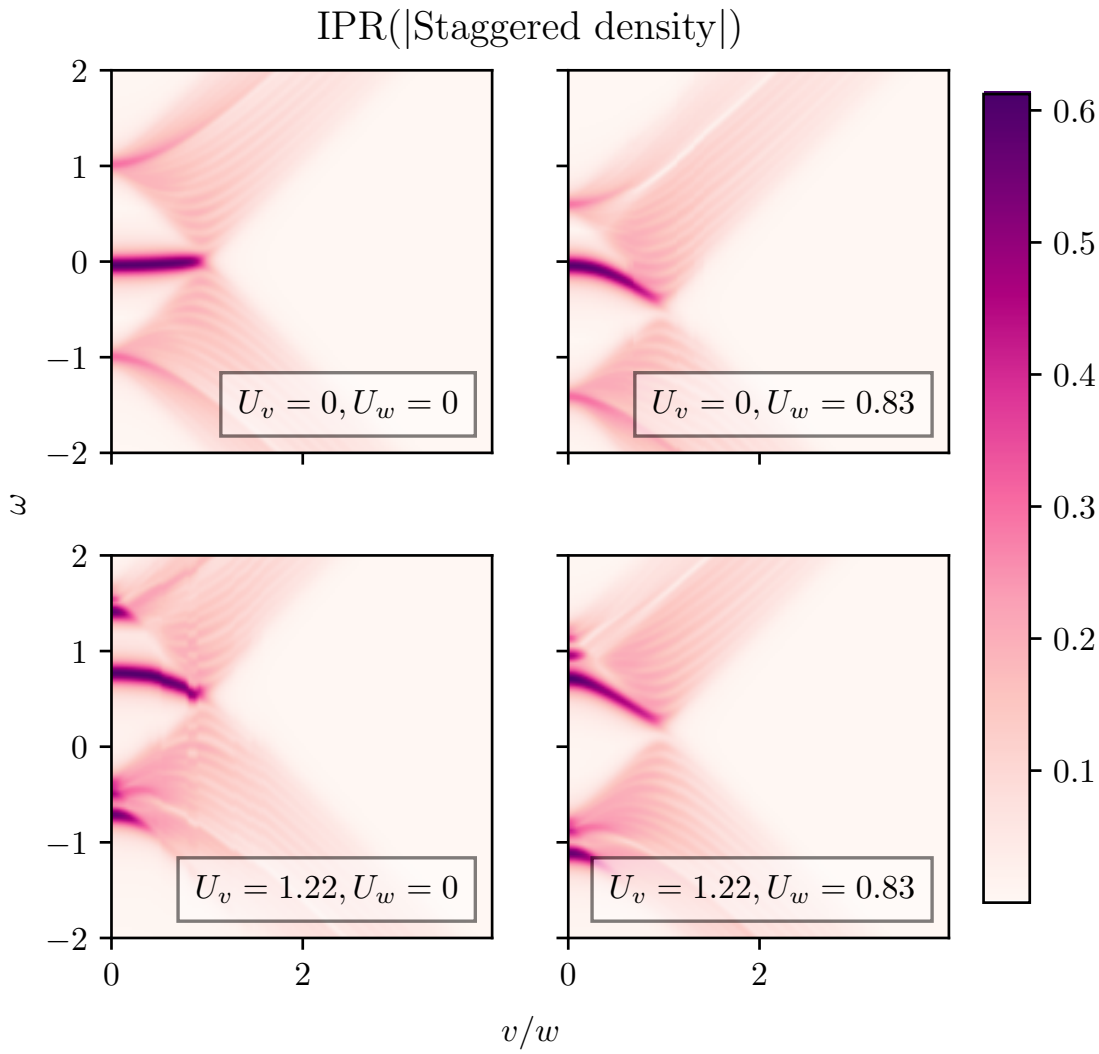

Figure 5.11: Inverse participation ratio of the staggered density, as a measure of correlation distance of the charge density wave. Phase diagram of the SSH model in the superconducting state with $\mu/w = 0$. The density of states are plotted as a function of the applied bias energy ω and intracell, interorbital hopping parameter v/w. A large impurity coupling $V = 20.28$ breaks the periodic boundary conditions.

Chapter 6

Model of $LaNiGa_2$

In this section we propose a tight-binding model of $LaNiGa_2$. We model only the interactions between Nickel atoms, as a minimal model of superconductivity. The Nickel atoms form a diatomic lattice, with two orbitals per site. We model in the plane such that the atoms are spatially degenerate. Therefore, the model is described by an onsite chemical potential tensor

$$\boldsymbol{\mu} \equiv \boldsymbol{\mu}_{\text{sites}} \otimes \boldsymbol{\mu}_{\text{spin}} \otimes \boldsymbol{\mu}_{\text{orbitals}}, \tag{6.1}$$

where

$$\boldsymbol{\mu}_{\text{sites}} \equiv \begin{pmatrix} -\mu_a & 0 \\ 0 & -\mu_b \end{pmatrix}, \tag{6.2}$$

$$\boldsymbol{\mu}_{\text{spin}} \equiv \mathbb{1}_2, \tag{6.3}$$

$$\boldsymbol{\mu}_{\text{orbitals}} \equiv \begin{pmatrix} 1+s & \delta \\ \delta & 1-s \end{pmatrix}, \tag{6.4}$$

and where μ_a and μ_b refer to the onsite chemical potentials for the atoms *above* and *below* the plane. The orbitals include a rigid shift s and hybridisation factor δ.

The intercell nearest-neighbour hopping is described by the tensor

$$\boldsymbol{T} \equiv \boldsymbol{T}_{\text{sites}} \otimes \boldsymbol{T}_{\text{spin}} \otimes \boldsymbol{T}_{\text{orbitals}}, \tag{6.5}$$

where

$$\boldsymbol{T}_{\text{sites}} \equiv \begin{pmatrix} -t & -t' \\ -t' & -t \end{pmatrix}, \tag{6.6}$$

$$\boldsymbol{T}_{\text{spin}} \equiv \mathbb{1}_2, \tag{6.7}$$

$$\boldsymbol{T}_{\text{orbitals}} \equiv \begin{pmatrix} 1+\alpha & 1+\beta \\ 1+\beta & 1-\alpha \end{pmatrix}, \tag{6.8}$$

and where $t > 0$ refers to intrasite and $t' < t$ refers to intersite tunnelling processes. $\alpha > 0$ describes intraorbital hopping ($d_{xy} - d_{xy}$, $d_{z^2} - d_{z^2}$) and $\beta > 0$ describes interorbital orbital hopping ($d_{xy} - d_{z^2}$, $d_{z^2} - d_{xy}$).

Chapter 7

Two-dimensional Su-Schrieffer-Heeger model with multiorbital superconductivity

As a first departure into the search for systems exhibiting topological superconductivity, we begin with a topologically non-trivial system. We will explore the effects of fermion pairing terms and of external magnetic fields on the system.

A canonical topological system is the Su-Schrieffer-Heeger (SSH) model of a spinless, one-dimensional dimer lattice of fermions with staggered hopping. The model obeys a *chiral symmetry*, exhibiting two topological phases. In the topologically non-trivial phase, a pair of edge states emerges at the boundaries, characterised by a *winding number*. In the interest of exploring two-dimensional superconductivity and the surface of three-dimensional superconductors, we study a two-dimensional extension of the SSH model. Bo-Hung extended the model in a manner which gives rise to topological modes. Their idea is to stack SSH models in a direct product with a one-dimensional Hamiltonian, known as a Type I extension. They refer to more complicated extensions as Type II.

7.1 One-dimensional SSH with nearest-neighbour intraorbital hopping

The simplest extension of the Su-Schrieffer-Heeger model with an additional intraorbital pairing parameter t is given by the following Hamiltonian

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \sum_{m\in\{A,B\}} \left[-\mu \hat{c}^\dagger_{x,m}\hat{c}_{x,m}(-t\hat{c}^\dagger_{x,m}\hat{c}_{x+1,m} + \text{h.c.})\right] + \sum_{x=1}^{n_x} \left[-v\hat{c}^\dagger_{x,A}\hat{c}_{x,B} - w\hat{c}^\dagger_{x,B}\hat{c}_{x+1,A} + \text{h.c.}\right] \quad (7.1)$$

We investigate the phases of the model in Figure **??**, and note that any nearest-neighbour intraorbital hopping is $t \neq 0$ gives rise to zero-energy states with density in the bulk. Indeed, the topological edge-states are a delicate situation. Mapping the SSH model onto a multiorbital model of superconducting, this shows that topological modes exist in a model in which the multiorbital-singlet term dominates over the conventional BCS singlet term, and in which the multiorbital-triplet term is negligible. In real materials, one would reasonably expect conventional BCS singlet pairing to dominate over multiorbital-singlet, and hence a mapping to the trivial phase of the SSH model.

7.2 Multiorbital tightbinding model, breaking chiral symmetry

An alternative model which breaks chiral symmetry is a multiorbital tightbinding chain with closed boundaries and an interorbital shift,

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \sum_{m\in\{A,B\}} \left[-(-1)^m \mu \hat{c}^\dagger_{x,m}\hat{c}_{x,m}\right] + \sum_{x=1}^{n_x} \left[-t\hat{c}^\dagger_{x,A}\hat{c}_{x,B} - t\hat{c}^\dagger_{x,B}\hat{c}_{x+1,A} + \text{h.c.}\right] \quad (7.2)$$

Although similar to the SSH model, the multiorbital shift model does not exhibit an entirely gapped bulk. As shown in the previous section, the fully bulk-gapped phase of the SSH model is extremely delicate and even the slightest intraorbital nearest-neighbour

../../05-fig/03-01-one-dimensional-ldos.pdf

Figure 7.1:

../../05-fig/03-02-multiorbital_chain.pdf

Figure 7.2:

perturbation gives rise to density in the bulk.

7.3 Spin-chain in an external magnetic field

We now consider a spin-chain, without orbital terms. Such a model would be realisable with an externally applied magnetic field, m. We suppose there is a spin-flip term, s, as well as a nearest-neighbour hopping term, t. The model is then described by,

$$\hat{\mathcal{H}}_{\text{SSH}} = \sum_{x=1}^{n_x} \left[+m\hat{c}^\dagger_{x,\uparrow}\hat{c}_{x,\uparrow} - m\hat{c}^\dagger_{x,\downarrow}\hat{c}_{x,\downarrow} \right] + \sum_{x=1}^{n_x} \left[s\hat{c}^\dagger_{x,\uparrow}\hat{c}_{x,\downarrow} \right] + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-t\hat{c}^\dagger_{x,\sigma}\hat{c}_{x+1,\sigma} + \text{h.c.} \right] \quad (7.3)$$

The advantage of this model is an obvious path to engineering: rather than orbitals, a spin-chain in an external magnetic field, which is a one-to-one mapping of the model.

7.4 Non-unitary spin-triplet Su-Schrieffer-Heeger model

The non-unitary spin-triplet extension of the Su-Schrieffer-Heeger model is given by

$$\hat{\mathcal{H}}_{\text{triplet SSH}} = \sum_{x=1}^{n_x} \sum_{m\in\{A,B\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-\mu\hat{c}^\dagger_{x,m,\sigma}\hat{c}_{x,m,\sigma}(-t\hat{c}^\dagger_{x,m,\sigma}\hat{c}_{x+1,m,\sigma} + \text{h.c.}) \right] + \sum_{x=1}^{n_x} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left[-v\hat{c}^\dagger_{x,A,\sigma}\hat{c}_{x,B,\sigma} - w\hat{c}^\dagger_{x,B,\sigma}\hat{c}_{x+1,A,\sigma} + \text{h.c.} \right] + \sum_{x=1}^{n_x} \left[-\sqrt{2}\Delta\hat{c}_{x,A,\uparrow}\hat{c}_{x,B,\uparrow} + \text{h.c.} \right] \quad (7.4)$$

The original SSH model is paramterised by vector

$$\mathbf{h}(k) = (v + w\cos(k), w\sin(k), 0). \quad (7.5)$$

Our extended model is parameterised in the following way

$$\hat{\mathcal{H}}_{\text{triplet SSH}} = \sigma_z \otimes \mathbb{1}_2 \otimes \mathbf{h}(k)\cdot\boldsymbol{\sigma} + (\sigma_+ \otimes i(\mathbf{d}\cdot\boldsymbol{\sigma})\sigma_y \otimes i\sigma_y + \text{h.c.}), \quad (7.6)$$

where the first term in the Kronecker product refers to the particle-hole channel, the second

corresponds to spin and the final sector is the orbital channel.

We remind the reader of the three basic symmetries of SU(4) matrices. The antiunitary symmetries are defined in terms of a complex conjugation operator **k** (in real space, so that the k-term flips sgn)

$$\mathbb{k}\hat{\mathcal{H}}\mathbb{k} = \hat{\mathcal{H}}^*(-k). \tag{7.7}$$

Our model breaks time-reversal symmetry (TRS). The corresonding operator t must satisfy the following equation

$$\mathbb{t}\hat{\mathcal{H}}\mathbb{t}^{-1} = -\hat{\mathcal{H}}^*(-k). \tag{7.8}$$

A tensor representation of the operator is

$$\mathbb{t} \equiv \mathbb{1}_2 \otimes i\sigma_y \otimes \mathbb{1}_2\mathbb{k} \tag{7.9}$$

which satisfies

$$\mathbb{t}^2 = -1 \tag{7.10}$$

.

The chiral symmetry $_z$ must satisfy the constraint

$$_z\hat{\mathcal{H}}_z^{-1} = -\hat{\mathcal{H}}^*(-k) \tag{7.11}$$

for which we find

$$_z = \mathbb{1}_2 \otimes \mathbb{1}_2 \otimes \sigma_z \tag{7.12}$$

and

$$(_z)^2 = +1. \tag{7.13}$$

Our model is a CII or DIII material in the classification of topological matter. In order to determine which, we must study the particle-hole symmetry operator $\mathbb{P}$, in which all terms acquire a sgn flip

$$\mathbb{P}\hat{\mathcal{H}}\mathbb{P}^{-1} = -\hat{\mathcal{H}}^*(-k). \tag{7.14}$$

A tensor representation for the corresponding operator is

$$\mathbb{P} = \sigma_x \otimes \mathbb{1}_2 \otimes \mathbb{1}_2\mathbb{k} \tag{7.15}$$

$$\mathbf{t}^2 = -1 \tag{7.16}$$

. Thus, our material falls into the symetry class DIII. In dimension $d = 1$ or $d = 2$, that is a $\mathbb{Z}_2$ topological number.

The characteristic equation is

$$|\mathbf{h}^{\cdot}$$

$\sigma - \lambda||\mathbf{h}^{\cdot}$
$\sigma + \lambda||(\mathbf{h}^{\cdot}$
$\sigma - \lambda)(\mathbf{h}^{\cdot}$
$\sigma + \lambda) + \Delta\Delta^{\dagger}|(7.17)$

7.5 Type I extension

7.5.1 Single interchain hopping

The simplest type I extension is s direct product of an SSH model, with hopping parameters v, w, with a tightbinding hamiltonian, with single interchain hopping parameter t, as depicted in Fig ref. The Hamiltonian is given by

$$\begin{aligned}\hat{\mathcal{H}}_{\text{SSH 2D}} = &\sum_{x=1}^{n_x}\sum_{y=1}^{n_y}\sum_{\sigma\in\{\uparrow,\downarrow\}}\sum_{m\in\{A,B\}}\left[-\mu\hat{c}^{\dagger}_{x,y,\sigma,m}\hat{c}_{x,y,\sigma,m}\right] \\ &+\sum_{x=1}^{n_x}\sum_{y=1}^{n_y}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left[-v\hat{c}^{\dagger}_{x,y,\sigma,A}\hat{c}_{x,y,\sigma,B} - w\hat{c}^{\dagger}_{x,y,\sigma,B}\hat{c}_{x+1,y,\sigma,A} + \text{h.c.}\right] \\ &+\sum_{m\in\{A,B\}}\left[-t\hat{c}^{\dagger}_{x,y,\sigma,m}\hat{c}_{x,y+1,\sigma,m} + \text{h.c.}\right],\end{aligned} \tag{7.18}$$

where the P-boundary is taken to be closed and the Q-boundary is open.

Chapter 8

Loop Josphson currents

Chapter 9

Hubbard U simulator: manual